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Semiglobal Safety-Filtered Extremum Seeking with Unknown CBFs

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Abstract— We introduce a safe extremum-seeking (Safe ES) algorithm which achieves the minimization of an unknown objective function while ensuring that an unknown, yet measured, control barrier function (CBF) remains above an arbitrarily small negative value for all time. In other words, "practical safety" is maintained during the entire period of convergence to the constrained extremum. Our design is based on quadratic program (QP) CBF style filters for safety, which is applied in an average and estimated sense. Using nonsmooth analysis tools, we guarantee semiglobal practical asymptotic (SPA) stability of the global constrained optimum, practical convergence to the safe set if starting in a condition violating the CBF, and practical safety for all time—semiglobally—if starting in safe set. The safety result of the paper is analogous with modern notions of SPA stability, guaranteeing that, for any small violation of safety, there exist design coefficients which guarantee that such a small violation is not exceeded. The paper outlines a set of sufficient conditions on the barrier and objective functions, and by way of a Lyapunov argument, we demonstrate that nonconvex constrained optimization problems can be solved. We present these results in the setting of a static map and a dynamical system. A simulation example illustrates the results.

Index Terms— Constrained Optimization, Extremum Seeking, Safe Control

I. Introduction

Given only measurements of the control barrier function (CBF) and without knowledge of the analytical form of the constraint, we present a constrained ES algorithm, which emulates the quadratic program (QP) CBF safety filters, despite the unavailable gradient of the CBF, and which minimizes an unknown but measured cost, while guaranteeing practical safety semiglobally. We further demonstrate through a Lyapunov argument that a class of nonconvex constrained optimization problems (where both the objective function and the constraint function may be nonconvex globally) can be solved under a suitable choice of design coefficients. The following results are achieved: 1) practical asymptotic stabilization—semiglobally, 2) practical convergence to the safe set—semiglobally, 3) practical safety for all time—semiglobally—if starting in the safe set.

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A. Literature

1) Extremum seeking: Extremum seeking (ES) is a powerful control method for dynamic systems with uncertainties. The first formal stability proof for ES feedback was given in [27]. Since then, various studies, extensions, and improvements of the ES algorithm have been made including the development of algorithms for semiglobal convergence [46], for maps with multiple extrema [48], extensions for multivariable maps [18], continuous and discrete stochastic generalizations [30], performance improvements for parameter convergence [2], [20], [34], for stabilizing open-loop unstable time-varying systems with unknown control directions [43], [44], for application to partial differential equations [35], for fixed-time Nash equilibrium seeking in time-varying networks [36], and for many other problems [40]. ES methods have also enabled the use of deep neural networks for time-varying systems without retraining [42], and have improved the robustness of generative convolutional neural networks beyond the span of their training data [41].

2) Constrained extremum seeking: ES has been studied in cases where constraints are unknown, often assuming convexity. This includes switching ES algorithms using projection maps [11], nonlinear programming techniques that smoothly switch between the gradient of the constraint and the objective [29], and solving convex problems using the saddle point of a "modified barrier function" [28]. The authors in [16], [17] also study saddle point dynamics and provide a Lie bracket-based analysis. In [7], the gradient of the objective is projected onto the gradient of the constraint. Additionally, various ES control laws have been studied where the constraint is known: simplex-like constraints on optimization parameters [37], constrained inputs to avoid actuator saturation [45], a switching framework where optimization parameters lie in a known set [38], and a form of ES with a bounded update law [44].

A number of notable data-sampled optimization schemes have been studied. Steady state constraint satisfaction is achieved in the framework of [21] as well as SPA stability, based on the data sampled schemes of [50]. Other data sampled frameworks are presented in [23] which study general classes of optimization schemes.

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3) CBFs and safety filters: The literature on safe control has significantly expanded recently due to the seminal work in [5], [53]. Many applications and other inspired works have emerged from QP-CBF designs, including incorporation with learning frameworks [15], [52], applications in robotic and automotive systems [39], [51], and designs involving observers under disturbances in dynamics and measurements [3], among others [10], [31].

Recent studies have explored input-to-state safety CBFs, where the violation of a safe set is bounded by the magnitude of disturbances [24]. Inverse optimal safety filters for disturbances of unlimited and unknown magnitude, both deterministic and stochastic, were introduced in [25]. Designs for CBFs of high relative degree were presented in [26], and high-order CBF conditions were studied in [22], [57]. Higher-order CBFs have been applied in optimal trajectory planning for autonomous vehicles [58]. Prescribed-time safety filters were designed in [1], and approximate optimal controllers, where safety violations are part of the value function, were formulated as an alternative to the QP-based approach [13].

B. Results and contributions of the paper

1) The goal of safety-filtered constrained ES and how it is achieved: This paper presents an ES algorithm designed to minimize an unknown objective function $J(\theta)$ over parameters θ, while ensuring system safety. Safety, represented by an unknown, yet measured, function $h(\theta)$, is maintained by keeping h positive.

Therefore, our algorithm tackles the following problem:

$$\min_{\theta} J(\theta) \text{ s.t. } h(\theta) \geq 0$$

where $h(\theta)$ and $J(\theta)$ are measured but analytically unknown with unknown gradients. The algorithm presented has the approximate characteristics:

1) if $h(\theta(t_0) \geq 0$ then $h(\theta(t)) \geq 0$ for all $t \in [0, \infty)$,
2) if $h(\theta(t_0)) < 0$ then $h(\theta(t)) \to h^+$ for some $h^+ \geq 0$ as $t \to \infty$,

where we allow initial conditions which may start unsafe – a situation not often considered in safety literature. Unsafe initial conditions approach safety, and safe initial conditions remain approximately safe for all time.

The classical unconstrained extremum seeking (ES) algorithm approximates the dynamics $\dot{\theta} = -\nabla J(\theta)$, performing gradient descent to minimize J. In the QP-CBF safety filter framework [6], $-\nabla J(\theta)$ serves as "nominal control," and a safety term u_s can be computed from $h(\theta)$, $\nabla h(\theta)$, and $\nabla J(\theta)$ [54]. The safety-filtered dynamics $\dot{\theta} = -\nabla J(\theta) + u_s$ satisfy $\dot{h}(\theta) + ch(\theta) \geq 0$, ensuring the safe set remains forward invariant. ES can estimate $\nabla J(\theta)$ and gradients of other unknown functions, allowing u_s to be applied to the ES dynamics. Thus, the Safe ES algorithm approximates $\dot{\theta} = -\nabla J(\theta) + u_s$, where u_s is an approximation of the QP-derived safety term. The block diagram of Safe ES is shown in Fig. 1.

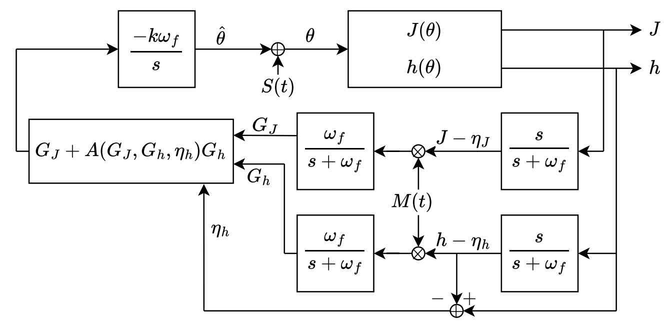

Fig. 1. Safe ES schematic – the quantity $\boldsymbol{A(G_J, G_h, \eta_h) = \min\{||G_h||^{-2}, M^+\} \max\{G_J^T G_h - c\eta_h, 0\} \geq 0}$ is a state dependent, scalar function.

2) Results of the paper: The framework from [33] and significant prior works [46], [47], [49] establishes semiglobal practical asymptotic (SPA) stability properties of extremum seeking. Using these results, we demonstrate that the Safe ES algorithm achieves SPA stability to the constrained optimum, and furthermore we demonstrate that "practical safety" may be achieved in addition to practical stability. Practical safety guarantees that for any violation of safety, however small, there exist design constants which can achieve it for all time. We make use of the two time scales built into the design, in which the estimated gradients converge fast, relative to the convergence of the parameter θ. Due to the approximation that ES provides, we do not expect to exactly solve the constrained optimization problem. Overall, the results are encompassed by the following three statements: 1) practical asymptotic stabilization—semiglobally 2) practical convergence to the safe set—semiglobally 3) practical safety for all time—semiglobally—if starting in the safe set. Furthermore, our conditions on J and h enable solving various nonconvex constrained optimization problems. We include an example at the end of this paper.

Safe ES allows the designer two degrees of freedom concerning safety: 1) the choice of the c 2) the design of h. Choosing c small slows the overall movement of the parameter towards the unsafe region when initialized in the safe set, and specifies the tradeoff between the favoring of safety and the favoring of the objective. The design of h may be derived from any signal such that a sufficient safety margin is already built in. For example, consider some measured signal $s(\theta(t)) : \mathbb{R}^n \mapsto \mathbb{R}$ which must never go below 0.5. Then one may conservatively design $h(\theta(t)) = s(\theta(t)) - 0.6$ if one is sure that the chosen perturbation signal and transient of the algorithm will not cause a large safety violation.

3) Contributions: We design an ES algorithm to solve constrained optimization problems where both the cost function and the constraint function are unknown but measurable. This is the first ES algorithm – to our knowledge – to 1) apply a QP-based safety filter with an unknown CBF and 2) solve nonconvex constrained optimization in continuous time. A Lyapunov argument demonstrating SPA stability of the controller to the constrained optimum reveals conditions on J and h that are more general than convexity.

Our main result is "practical safety," analogous to "practical stability" in modern ES schemes. For an arbitrarily small safety violation, there exist design constants that ensure system trajectories do not exceed that violation if they start in the safe set. This guarantee of preventing violation (in a practical sense) for all time, rather than just asymptotically, leverages the two time scales in the algorithm. Additionally, we allow initial conditions far outside the safe set and guarantee practical convergence to the safe set semiglobally.

4) Added value over conference versions: We expand upon the conference presentation [55], which only included theorem statements due to page limits. Here, we present the full results, including proofs using nonsmooth tools, and introduce Safe ES results for dynamical systems. We also further elaborate on practical safety. The first ever attempt to incorporate a CBF-QP safety filter with ES, offering only a local analysis of a scalar parameter under restrictive assumptions on J and h, is detailed in [54]. Safe ES was also recently implemented in applications to particle accelerator systems [56].

5) Notation: We denote $\mathbb{R}_{>0}$ and $\mathbb{R}_{\geq 0}$ as the set of positive and non-negative real numbers, respectively. For a differentiable function $Q : \mathbb{R}^n \to \mathbb{R}$ we denote the gradient as $\nabla Q : \mathbb{R}^n \to \mathbb{R}^n$. For a vector $v \in \mathbb{R}^n$, the notation $\|v\|$ denotes the Euclidean norm. The continuous function $\beta : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is of class $\mathcal{K}$ if $\beta(0) = 0$ and it is strictly increasing. The continuous function $\beta : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is of class $\mathcal{KL}$ if it is strictly increasing in its first argument and strictly decreasing to zero in its second argument. The image of a function h is denoted by $\mathrm{Im}(h)$. The open ball around a point p is $B_r(p) = \{\theta \in \mathbb{R}^n : \|\theta - p\| < r\}$ and the closed ball is $\bar{B}_r(p)$. For some nonzero vector v we may denote the normalized vector as $\hat{v} = v/\|v\|$. The boundary and interior of a set $\mathcal{C}$ are denoted as $\mathrm{bd}(\mathcal{C})$ and $\mathrm{Int}(\mathcal{C})$ respectively. We use the term "SPA stability" to refer to the notion of semiglobal practical asymptotic stability [47]. A function $f(x, \epsilon)$ is $O(\epsilon)$ if for any compact set Ω there exists a positive pair (ϵ^*, k) such that $\|f(x, \epsilon)\| \leq k\epsilon$ for all $\epsilon \in (0, \epsilon^*]$ for all $x \in \Omega$. The convex hull of a set $\mathcal{S}$ is $\mathrm{co}(\mathcal{S})$. Given $a \in \mathbb{R}^n$ and $b \in \mathbb{R}^m$ we often use the notation $u = (a, b) = [a^T, b^T]^T \in \mathbb{R}^{n+m}$ when stacking vectors, and a slight abuse of notation when indexing components of a stacked vector so that $u_1 = a \in \mathbb{R}^n$ and $u_2 = b \in \mathbb{R}^m$.

II. Assumptions

We define

$$\mathcal{C} = \{\theta \in \mathbb{R}^n : h(\theta) \geq 0\} \tag{1}$$

called the 'safe set'. We also define the notation of a superlevel set of h, parameterized by the parameter $\rho \leq 0$ as

$$\mathcal{C}_\rho = \{\theta \in \mathbb{R}^n : h(\theta) \geq \rho, \rho \in \mathrm{Im}(h) \cap \mathbb{R}_{\leq 0}\}. \tag{2}$$

The sets of the form $\mathcal{C}_\rho$ always contain $\mathcal{C}$ along with some unsafe region given by ρ, a non-positive value in the image of h. We also use the following assumptions throughout.

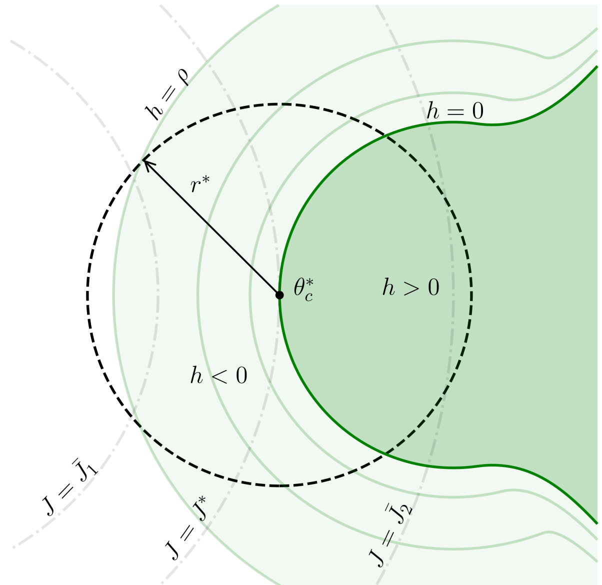

Fig. 2. Depiction of sets and levels of $\boldsymbol{h}$ and $\boldsymbol{J}$ with $\bar{\boldsymbol{J}}_1 < \boldsymbol{J}^* < \bar{\boldsymbol{J}}_2$. The safe set $\boldsymbol{\mathcal{C}}$ is shaded in dark green, which is contained by the set $\boldsymbol{\mathcal{C}}_{\boldsymbol{\rho}}$ defined by both dark and light green shading. The radius $\boldsymbol{r}^*$ relates to the "angle condition" given in Assumption 4.

Assumption 1 (Objective Function Conditions): The objective function $J : \mathbb{R}^n \to \mathbb{R}$ is differentiable with locally Lipschitz Jacobian and satisfies:

1) $\theta_c^* \in \mathcal{C}$ is the unique constrained minimizer of J on $\mathcal{C}$ denoted as $J^* = J(\theta_c^*)$,
2) if there exists a θ such that $\nabla J(\theta) = 0$ for $\theta \in \mathcal{C}$, then $\theta = \theta_c^*$.

Assumption 2 (Barrier Function Conditions): The barrier (or safety) function $h : \mathbb{R}^n \to \mathbb{R}$ is differentiable with locally Lipschitz Jacobian and satisfies:

1) the safe set $\mathcal{C}$ is non-empty,
2) for any $\mathcal{C}_\rho$, there exists a $L \in (0, \infty)$ such that $\|\nabla h(\theta)\| > L$ for $\theta \in \{h(\theta) \leq 0\} \cap \mathcal{C}_\rho$.

Assumption 3 (Optimizer Condition): If $\nabla h(\theta)^T \nabla J(\theta) = \|\nabla h(\theta)\| \; \|\nabla J(\theta)\|$ ($\nabla h(\theta)$ and $\nabla J(\theta)$ are collinear) for $\theta \in \mathrm{bd}(\mathcal{C})$, then $\theta = \theta_c^*$.

Assumption 4 (Angle Condition): For any $\rho \in \mathrm{Im}(h) \cap \mathbb{R}_{<0}$, there exists a $r^* > 0$ and $f^* \in [0, 1)$ such that

$$\frac{\nabla J(\theta)^T \nabla h(\theta)}{\|\nabla J(\theta)\| \; \|\nabla h(\theta)\|} \leq f^*, \tag{3}$$

for $\theta \in \{\rho \leq h(\theta) \leq 0\}$, $\theta \notin B_{r^*}(\theta_c^*)$.

Assumption 5 (Radial Unboundedness): The function $V(\theta) = \max\{-h(\theta), 0\} + \max\{J(\theta) - J^*, 0\}$ is positive definite and $\|\theta - \theta_c^*\| \to \infty \implies V \to \infty$.

Assumption 6 (Bounded Levels of h): All sets $\mathcal{C}_\rho$ are compact.

Assumption 7 (ES Constants): The design constants are chosen as $\omega_\mathrm{f}, \omega_i, \delta, a, k, c > 0$, where ω_i/ω_j are rational with frequencies ω_i chosen such that $\omega_i \neq \omega_j$ and $\omega_i + \omega_j \neq \omega_k$ for distinct i, j, and k.

In our analysis we use Assumptions 1, 2, and 3 to establish a unique equilibrium in the dynamics of a so

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called 'exact system', which our ES scheme approximates. This equilibrium is identically the constrained optimizer θ_c^*. Assumptions 4, 5, and 6 facilitate the subsequent Lyapunov analysis.

Assumption 3 ensures a single equilibrium by imposing constraints on the gradients of J and h at their boundary. This condition aligns with principles from optimization theory. The gradient condition is also apparent in Fig. 2 in the tangency of the levels of J with $h = 0$.

Assumption 4 focuses on the cosine of the angle between $\nabla J(\theta)$ and $\nabla h(\theta)$ far away from the constrained equilibrium, but only in the unsafe set, $\theta \in \{\rho \leq h(\theta) \leq 0\}$. It aids in deriving a Lyapunov function with a universally negative time derivative, applicable, for instance, in scenarios with semi-infinite safe sets, as depicted in Fig. 2. Proposition 1 illustrates that for a large class of problems, quadratic and positive definite J with linear h, Assumption 4 holds. The proof is given in the appendix.

Proposition 1: If J is a quadratic, positive definite function and h is a linear function satisfying Assumption 2, then Assumption 4 is satisfied.

Assumption 5 ensures that the objective function does not asymptotically plateau as θ extends away from the equilibrium in safe directions. Our main results hinge upon the satisfaction of either Assumptions 4-5 or Assumption 6. The latter, asserting all sets $\mathcal{C}_\rho$ are compact, exhibits sufficient strength to establish Semiglobal Practical Asymptotic (SPA) stability in the reduced system. However, for scenarios with semi-infinite safe sets, the Assumptions 4-5 become necessary for our main results to hold.

The assumptions listed here a minimal set of a nonrestrictive conditions we have gathered which guarantee the main results of the paper. There are indeed slightly less restrictive assumptions that may be taken instead, relaxing some of the conditions. For example, Assumptions 4 is truly only necessary if some $\mathcal{C}_\rho$ is an unbounded set and $\|\nabla J(\theta)\|$ grows unbounded on $\mathcal{C}_\rho$. If $\|\nabla J(\theta)\|$ does not grow unbounded on this $\mathcal{C}_\rho$ (a set which can be bounded or not), Assumption 4 is unnecessary.

III. Preliminaries on nonsmooth Lyapunov Analysis

In this section we introduce concepts of regularity, the set-valued Lie derivative, and stability theorems. Nonsmooth analysis tools are necessary as we use the Lyapunov function $V_\alpha(\theta) = \max\{-\alpha h(\theta), 0\} + \max\{J(\theta) - J^*, 0\}$ for the main stability results of the paper.

A. The Generalized Gradient and Set-Valued Lie Derivative

Definition 1: Let $V : \mathbb{R}^n \to \mathbb{R}$ be locally Lipschitz and let Ω_V be a set of measure zero containing the points where V is not differentiable. The generalized gradient is defined as

$$\partial V(x) = \mathrm{co}\left\{\lim_{i\to\infty} \nabla V(x_i) : x_i \to x, x_i \notin \Omega_V\right\}. \quad (4)$$

The definition of the generalized gradient states that if we consider any sequence x_i converging to x such that it avoids a measure zero set where $V(x_i)$ is not differentiable; then the convex combination of all such sequences of $\nabla V(x_i)$ is $\partial V(x)$. Recall that Rademacher's Theorem states if V is Lipschitz then it is differentiable almost everywhere. See [12, Theorem 2.5.1] and [14, Equation 37] for more information on generalized gradients.

Regularity of a Lyapunov function proves to be very useful in calculating generalized derivatives. The classical definition of regularity can be seen in [12], but the following known class of regular functions is sufficient background for this paper, rewritten from [12, Proposition 2.3.12].

Proposition 2: For a finite collection of functions ($k = 1, 2, \ldots, m$) let $g_k : \mathbb{R}^n \to \mathbb{R}$ be differentiable (or more generally regular and locally Lipschitz) functions, and let $V : \mathbb{R}^n \to \mathbb{R}$ be defined as the maximum of the collection of functions

$$V(x) = \max\{g_k(x) : k = 1, 2, \ldots, m\}. \quad (5)$$

Then:

1) V is regular and locally Lipschitz.
2) Let $I_V(x)$ denote the set of indices k for which $V(x) = g_k(x)$, then

$$\partial V(x) = \mathrm{co}\left\{\nabla g_i(x) : i \in I_V(x)\right\}. \quad (6)$$

The generalized gradient has been defined and a rule given to calculate it (6) for functions of the form (5).

We now define a set-valued form of the Lie derivative, utilizing the generalized gradient [14]. It can be more generally defined for a differential inclusion, but we have defined it below for a differential equation,

$$\dot{x} = f(x) \quad (7)$$

with $f : \mathbb{R}^n \to \mathbb{R}^n$ locally Lipschitz.

Definition 2: Let $V : \mathbb{R}^n \to \mathbb{R}$ be locally Lipschitz. The set-valued Lie derivative of V with respect to f in (7) is defined as

$$\mathcal{L}_f V(x) = \{a \in \mathbb{R} : f(x)^T p = a \text{ for all } p \in \partial V(x)\}. \quad (8)$$

If $V(x)$ is differentiable at x, then the usual time derivative coincides with the set-valued Lie derivative $\dot{V} = \mathcal{L}_f V(x) = \{f(x)^T \nabla V(x)\}$. However, if $V(x)$ is not differentiable at x then $\mathcal{L}_f V(x)$ will represent the convex hull of the set of all possible time derivatives obtained at x when approaching x from all possible directions as defined by the generalized gradient.

B. Stability and Invariance Results

The next two stability results are nonsmooth versions of Lasalle's invariance principle and Nagumo's theorem. The following result is based on [8], [14].

Theorem 1: Let $V : \mathbb{R}^n \to \mathbb{R}$ be locally Lipschitz and regular. Let $\mathcal{S} \subset \mathbb{R}^n$ be compact and positively invariant for (7). Suppose $\max L_f V(x) \leq 0$ for all $x \in \mathcal{S}$. Then all solutions of (7), with $x(t_0) \in \mathcal{S}$, converge to the largest invariant set $\mathcal{M}$ contained in

$$\mathcal{S} \cap \overline{\{x \in \mathbb{R}^n : 0 \in L_f V(x)\}}. \tag{9}$$

The next result is a statement of Nagumo's theorem [32], more simply reformulated from [9, Section 4.2].

Theorem 2: Suppose $\mathcal{S} \subseteq \mathbb{R}^n$ is a closed set. Then $\mathcal{S}$ is invariant with respect to (7) if and only if

$$f(x) \in \mathcal{T}_{\mathcal{S}}(x) \text{ for all } x \in \text{bd}(\mathcal{S}), \tag{10}$$

where $\mathcal{T}_{\mathcal{S}}(x)$ is Bouligand's tangent cone defined as

$$\mathcal{T}_{\mathcal{S}}(x) = \left\{ z : \liminf_{\tau \to 0} \frac{\text{dist}(x + \tau z, \mathcal{S})}{\tau} = 0 \right\} \tag{11}$$

with the distance from a set defined as $\text{dist}(y, \mathcal{S}) = \inf_{w \in \mathcal{S}} \|y - w\|$ for some $y \in \mathbb{R}^n$.

The tangent cone $\mathcal{T}_{\mathcal{S}}(x)$ essentially describes the set of vectors pointing tangent along the boundary of $\mathcal{S}$ or "inward". For example, consider $\mathcal{C} = \{h(\theta) \geq 0\}$, for h with properties given in Assumption 2. Then for $\theta \in \text{bd}(\mathcal{C})$, $\mathcal{T}_{\mathcal{C}}(\theta) = \{z : \nabla h(\theta)^T z \geq 0\}$.

The following lemma makes clear many of the assumptions and restrictions on J and h we have considered in this paper.

Lemma 1: Consider the locally Lipschitz function $V_\alpha : \mathbb{R}^n \to \mathbb{R}$ defined as

$$V_\alpha(\theta) = \max\{-\alpha h(\theta), 0\} + \max\{J(\theta) - J^*, 0\}, \tag{12}$$

for some parameter $\alpha > 0$ under the Assumptions 1 - 3. Then

1) V_α is regular and can be written as

$$V_\alpha(\theta) = \max\left\{-\alpha h(\theta), J(\theta) - J^*, -\alpha h(\theta) + J(\theta) - J^*\right\}. \tag{13}$$

2) Let either Assumption 4 - 5 hold, or 6 hold. Then for any $\rho < 0$, there is a sufficiently large $\alpha > 0$ such that the following holds for the set $\mathcal{R} = \{V_\alpha(\theta) \leq \bar{V}\} \cap \mathcal{C}_\rho$ for any $\bar{V} > 0$:
   a) the tangent cone $\mathcal{T}_{\mathcal{R}}(\theta)$ is:
   $$\mathcal{T}_{\mathcal{R}}(\theta) = \{v : \max\{\xi^T v : \xi \in \partial W(\theta)\} \leq 0\}, \tag{14}$$
   for $\theta \neq \theta_c^*$, with
   $$\begin{aligned} W(\theta) = \max\{ & -h(\theta) + \max\{\rho, -\bar{V}/\alpha\}, \\ & J(\theta) - J^* - \bar{V}, \\ & J(\theta) - J^* - \bar{V} - \alpha h(\theta)\}. \end{aligned} \tag{15}$$
   b) the set $\mathcal{R} = \{\theta : W(\theta) \leq 0\}$ is compact and $\{W(\theta) = 0\} \equiv \text{bd}(\mathcal{R})$.
   c) for $W(\theta) = 0$, $0 \notin \partial W(\theta)$.

The above lemma shows the regularity of V_α, under the assumed properties of J and h, when α is chosen sufficiently large. It also provides simple expressions for tangent cones on sets involving the levels of V_α using mainly the results of [12], and useful properties of these sets.

IV. Global Convergence of the Exact Algorithm

Before conducting analysis of the ES scheme for unknown functions, we study the optimization algorithm in its exact form in which we assume that all gradients are known, in order to find the appropriate Lyapunov function which will be used in the next section. Consider the following dynamics:

$$\begin{aligned} \dot{\theta} = F(\theta) = & -\nabla J(\theta) + \\ & \frac{\nabla h(\theta)}{\|\nabla h(\theta)\|^2} \max\{\nabla J(\theta)^T \nabla h(\theta) - ch(\theta), 0\}. \end{aligned} \tag{16}$$

Note that there is no risk of dividing by zero in the expression of (16) as $\nabla h(\theta) = 0 \implies h(\theta) > 0$ by Assumption 2. Therefore, $\lim_{\|\nabla h(\theta)\| \to 0} F(\theta) = -\nabla J(\theta)$ on compact sets.

The differential inequality $\dot{h} + ch \geq 0$ is commonly used to show the forward invariance of the safe set [6], where it is assumed that the initial condition of any given trajectory is safe. But it also shows attractivity to the safe set in the case where $h(\theta(t)) < 0$, for some t, implying that $\dot{h}(\theta(t)) > 0$ and therefore h is increasing. So unsafe trajectories become 'more safe' in an exponential fashion. Therefore, the family of sets $\mathcal{C}_\rho$ is also positively invariant, and not just the case of $\rho = 0$. We state this formally below.

Proposition 3: Under Assumption 2, the dynamics (16) satisfy $\frac{dh(\theta(t))}{dt} + ch(\theta(t)) \geq 0$ for all $\theta \in \mathbb{R}^n$ and $\mathcal{C}$ is forward invariant. Moreover, all sets $\mathcal{C}_\rho$ are forward invariant and $h(\theta(t)) \geq h(\theta(t_0))e^{-ct}$ for all $\theta(t_0) \in \mathbb{R}^n$.

Proof: We can express the exact dynamics as (16) under Assumption 2 and simply compute

$$\begin{aligned} \frac{dh(\theta(t))}{dt} + ch(\theta) = & -\nabla J(\theta)^T \nabla h(\theta) + ch(\theta) + \\ & \max\{\nabla J(\theta)^T \nabla h(\theta) - ch(\theta), 0\} \geq 0 \end{aligned} \tag{17}$$

where we use the fact that $-x + \max\{x, 0\} \geq 0$ for any $x \in \mathbb{R}$. By the comparison principle, $h(\theta(t)) \geq h(\theta(t_0))e^{-ct}$ for all $\theta(t_0) \in \mathbb{R}^n$ and all sets $\mathcal{C}_\rho$ are forward invariant. ■

This proposition will help us later establish global convergence of the algorithm for trajectories starting outside of $\mathcal{C}$.

We now discuss the equilibrium of (16). The previous proposition has guaranteed that the equilibrium of the algorithm cannot lie in an unsafe region, as $\dot{h}(\theta) > 0$ for $h(\theta) < 0$. Therefore, consider solving for $F(\theta^e) = 0$ for $h(\theta^e) \geq 0$ under Assumptions 1, 2, and 3. Clearly if $\nabla J(\theta^e) = 0$ then $\theta^e = \theta_c^*$ by Assumption 1. If $\nabla J(\theta^e) \neq 0$ for $h(\theta^e) \geq 0$, computing the inner product $\nabla J(\theta^e)^T F(\theta^e)$ shows that $F(\theta^e) = 0$ only when the following two things are true at the same time: 1) $h(\theta^e) = 0$ and 2) $\nabla h(\theta^e)$ and $\nabla J(\theta^e)$ are collinear. Then, by Assumption 3 this implies $\theta^e = \theta_c^*$. All in all, we have that

$$F(\theta^e) = 0 \implies \theta^e = \theta_c^*. \tag{18}$$

It turns out that the inner product $\nabla J(\theta)^T F(\theta)$ is also always negative for $h(\theta) \geq 0$ and $\theta \neq \theta_c^*$. Therefore we can formulate the Lyapunov function $V_1(\theta) = J(\theta) - J(\theta^e)$

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in order to prove stability for $\theta \in \mathcal{C}$. Computing $\dot{V}_1 = \nabla J(\theta)^T F(\theta)$ we have,

$$\dot{V}_1 = -\|\nabla J(\theta)\|^2 + \frac{\nabla J(\theta)^T \nabla h(\theta)}{\|\nabla h(\theta)\|^2} \max\{\nabla J(\theta)^T \nabla h(\theta) - ch(\theta), 0\} \quad (19)$$

which is clearly negative definite if $\nabla J(\theta)^T \nabla h(\theta) - ch(\theta) \leq 0$. If not,

$$\dot{V}_1 = -\nabla J(\theta)^T \left(I - \frac{\nabla h(\theta) \nabla h(\theta)^T}{\|\nabla h(\theta)\|^2} \right) \nabla J(\theta) - c\frac{\nabla J(\theta)^T \nabla h(\theta)}{\|\nabla h(\theta)\|^2} h(\theta). \quad (20)$$

The quadratic first term is negative (unless $\theta = \theta_c^*$) and the second term is negative semidefinite on $\theta \in \{h(\theta) \geq 0\} \cup \{\nabla J(\theta)^T \nabla h(\theta) - ch(\theta) > 0\}$, $\theta \neq \theta_c^*$. Because the set $\mathcal{C} := \{h(\theta) \geq 0\}$ is forward invariant, and V_1 is a smooth Lyapunov function on $\mathcal{C}$, we have the following Lemma.

Lemma 2: Let Assumptions 1-3 hold. The function $V_1(\theta) = J(\theta) - J(\theta^e)$ is a Lyapunov function for the equilibrium θ_c^* on $\mathcal{C}$, yielding strictly $\dot{V}_1 < 0$ for all $\theta \in \mathcal{C} \setminus \{\theta_c^*\}$. The dynamics (16) are asymptotically stable to the constrained minimizer θ_c^* for $\theta(t_0) \in \mathcal{C}$.

Proof: Step 1: construct a forward invariant set $\mathcal{R} = \{V_1(\theta) \leq \bar{V}\} \cap \mathcal{C}$ for any $\bar{V} > 0$. We represent this set as $\mathcal{R} = \{W(\theta) \leq 0\}$ with $W(\theta) = \max\{V_1(\theta) - \bar{V}, -h(\theta)\}$. The set is compact, and can be shown to be forward invariant with Theorem 2 by showing $F(\theta) \in \mathcal{T}_R(\theta)$. The tangent cone can be expressed using generalized gradients of W:

$$\mathcal{T}_{\mathcal{R}}(\theta) = \{v : \max\{\xi^T v : \xi \in \partial W(\theta)\} \leq 0\}, \quad (21)$$

For details showing that $\mathcal{R}$ has the regularity properties necessary to prove the statement above, see the arguments in the proof of Statement 2a, Lemma 1 - we omit demonstrating these regularity properties due to space. Now, check the negativity of $F(\theta)^T p$ where $p \in \partial W(\theta)$ for $\theta \in \mathrm{bd}(\mathcal{R})$. If $\theta \notin \{h(\theta) = 0\}$ then $F(\theta)^T p = \dot{V}_1 < 0$, as shown in (19) and (20). If $\theta \in \{h(\theta) = 0\}$ then $p = -\lambda_1 \nabla h(\theta) + \lambda_2 \nabla J(\theta)$, for $\lambda_1 + \lambda_2 = 0, \lambda_1 \geq 0, \lambda_2 \geq 0$. Then $F(\theta)^T p = -\lambda_1 \dot{h}(\theta) + \lambda_2 \dot{V}_1(\theta) < 0$ by Proposition 3 and using the fact that $\dot{V}_1(\theta) < 0$. By Theorem 2, $\mathcal{R}$ is forward invariant

Step 2: apply Theorem 1. The result holds because $\dot{V}_1(\theta) < 0$ on the compact invariant set $\mathcal{R} \setminus \{\theta_c^*\}$. ■

Because Theorem 1 requires a compact invariant set, the proof above uses the boundary of the safe set (a potentially unbounded set) and a level of the Lyapunov function $V_1 = \bar{V}$ in order to construct this compact invariant set. This idea is used again in the next lemma where we use the boundary $h = \rho$ of a set $\mathcal{C}_\rho$ and the levels of a different Lyapunov function V_α.

In light of Proposition 3 (attractivity to $\mathcal{C}$) and Lemma 2 (convergence within $\mathcal{C}$), it should be intuitive to the reader that all trajectories eventually converge on the equilibrium point θ_c^*. We can demonstrate this fact with the following Lyapunov argument on any invariant set in Lemma 3.

We use Theorem 1 in the following lemma to prove global asymptotic stability. In order to use Theorem 1 we first construct a compact invariant set $\mathcal{R}$ using the intersection of $\mathcal{C}_\rho$ and a sublevel set of the Lyapunov function V_α. A function W is used to define the set $\mathcal{R}$ and check that the dynamics $F(\theta)$ point 'inward' to $\mathcal{R}$ at it's boundary. The boundary is not smooth, which is why generalized gradients are key tool. Based on the assumptions on the functions J and h, an essential idea in achieving global stability is the fact that we can design the set $\mathcal{R}$ to be arbitrarily large, encompassing any initial condition.

Lemma 3: Let Assumptions 1-3 hold, and either Assumptions 4-5 or Assumption 6. Consider the Lypaunov function

$$V_\alpha(\theta) = \max\{-\alpha h(\theta), 0\} + \max\{J(\theta) - J(\theta_c^*), 0\}. \quad (22)$$

For any invariant set $\mathcal{C}_\rho$, there exists $\alpha \in (0, \infty)$ such that $\max L_f V_\alpha(x) < 0$ for $\theta \in \mathcal{C}_\rho \setminus \{\theta_c^*\}$. The dynamics (16) are globally asymptotically stable to the constrained minimizer θ_c^*.

Proof: Step 1: show that for any set $\mathcal{C}_\rho$ there is a sufficiently large $\alpha > 0$ such that $\mathcal{R} = \{V_\alpha(\theta) \leq \bar{V}\} \cap \mathcal{C}_\rho$ is compact and invariant for any $0 < \bar{V} < \infty$. From Lemma 1, V_α is regular and $\mathcal{R}$ is compact, therefore we can determine if $F(\theta)$ lies in $\mathcal{T}_{\mathcal{R}}(\theta)$ by showing that $F(\theta)^T p < 0$ for $p \in \partial W(\theta)$ and $\theta \in \{W(\theta) = 0\} \setminus \{\theta_c^*\}$ with

$$\begin{aligned} W(\theta) = \max\{ & -h(\theta) + \max\{\rho, -\bar{V}/\alpha\}, \\ & J(\theta) - J^* - \bar{V}, \\ & J(\theta) - J^* - \bar{V} - \alpha h(\theta)\}. \end{aligned} \quad (23)$$

Also in Lemma 1 we showed that for sufficiently large $\alpha > 0$, $p \neq 0$ for $W(\theta) = 0$. Therefore we assume, by default, in the following cases of h that 1) $\alpha > 0$ is sufficiently large based on any fixed $\rho < 0$ and 2) $\theta \in \{W(\theta) = 0\} \setminus \{\theta_c^*\}$.

Case 1): $h(\theta) > 0$. Then $W(\theta) = J(\theta) - J^* - \bar{V}$ and $\partial W(\theta) = \{\nabla J(\theta)\} \neq \{0\}$. With $p = \nabla J(\theta)$ it is clear from (19) and (20) that $\dot{V}_1 = F(\theta)^T p < 0$ for any $\alpha > 0$.

Case 2): $h(\theta) = 0$. Then $\partial W(\theta) = \mathrm{co}\{\nabla J(\theta), \nabla J(\theta) - \alpha \nabla h(\theta)\}$. For some $p \in \partial W(\theta)$, $p = \nabla J(\theta) - \alpha\lambda \nabla h(\theta)$, for $0 \leq \lambda \leq 1$. If $\nabla J(\theta)^T \nabla h(\theta) \leq 0$ then

$$F(\theta)^T p = -\|\nabla J(\theta)\|^2 + \alpha\lambda \nabla J(\theta)^T \nabla h(\theta) < 0 \quad (24)$$

for any $\alpha > 0$.

If $\nabla J(\theta)^T \nabla h(\theta) > 0$ then by Assumption 3,

$$\begin{aligned} F(\theta)^T p = \nabla J(\theta)^T \left(\frac{\nabla h(\theta)^T \nabla h(\theta)}{\|\nabla h(\theta)\|^2} - I \right) \nabla J(\theta) \\ - \alpha\lambda \nabla J(\theta)^T \nabla h(\theta) < 0 \end{aligned} \quad (25)$$

for any $\alpha > 0$.

Case 3): $h(\theta) < 0$. So $\partial W(\theta) = \mathrm{co}\{-\nabla h(\theta), \nabla J(\theta) - \alpha \nabla h(\theta)\}$. For some $p \in \partial W(\theta)$, we have $p = -\lambda_1 \nabla h(\theta) + \lambda_2(-\alpha \nabla h(\theta) + \nabla J(\theta))$, with $0 \leq \lambda_1, \lambda_2 \leq 1$ and $\lambda_1 + \lambda_2 = 1$.

We define the lumped quantities

$$m(\theta) = \nabla h(\theta)^T \nabla J(\theta) - ch(\theta), \quad (26)$$

$$f(\theta) = \nabla J(\theta)^T \nabla h(\theta) / (\|\nabla J(\theta)\| \|\nabla h(\theta)\|) \leq 1. \quad (27)$$

If θ lies in a region where $m(\theta) \le 0$ then $\nabla J(\theta)^T h(\theta) \le ch(\theta) < 0$ and one can easily show $F(\theta)^T p < 0$. Otherwise, if $m(\theta) > 0$ then

$$F(\theta)^T p = c|h(\theta)| \left(-\lambda_1 + \lambda_2 \left(-\alpha + f(\theta) \frac{\|\nabla J(\theta)\|}{\|\nabla h(\theta)\|} \right) \right) - \lambda_2 \|\nabla J(\theta)\|^2 \left(1 - f^2(\theta)\right). \quad (28)$$

We consider θ to lie in a region such that $m(\theta) > 0$ in the following two sub-cases.

Case 3a): Assumption 6 holds. Then set $\mathcal{C}_\rho$ is compact so choose

$$\alpha > \sup_{\theta \in \mathcal{C}_\rho} \frac{\|\nabla J(\theta)\|}{L}, \quad (29)$$

using Assumption 2 (which states $\|\nabla h(\theta)\| > L > 0$ for $h(\theta) < 0$). This yields $F(\theta)^T p < 0$ for all $\theta \in \mathcal{C}_\rho \setminus \{\theta_c^*\}$, therefore $\mathcal{R}$ is forward invariant by Theorem 2. Note that this argument also holds if $\|\nabla J(\theta)\|$ is bounded on a potentially unbounded set $\mathcal{C}_\rho$.

Case 3b): Assumptions 4 - 5 hold. Assume the non trivial scenario where $\|\nabla J(\theta)\|$ is unbounded on $\mathcal{C}_\rho$. Then from Assumption 4, there exists a scalar f^* and set $B_{r^*}(\theta_c^*)$ such that $0 < f(\theta) \le f^* < 1$ for all $\theta \notin B_{r^*}(\theta_c^*)$. Let $\tilde{f} = 1 - {f^*}^2$ with $\tilde{f} \in (0, 1)$. Then for any $\theta \notin B_{r^*}(\theta_c^*)$,

$$F(\theta)^T p = -c|h(\theta)|(\lambda_1 + \alpha\lambda_2) + \lambda_2 \|\nabla J(\theta)\| \left(\frac{c|\rho|}{L} - \tilde{f} \|\nabla J(\theta)\| \right), \quad (30)$$

using Assumption 2. From the expression in the equation above, $\|\nabla J(\theta)\| > |\rho|c/L\tilde{f}$ implies $F(\theta)^T p < 0$ for $\theta \notin B_{r^*}(\theta_c^*)$ regardless of α. Therefore, consider $\|\nabla J(\theta)\| \le |\rho|c/L\tilde{f}$. From (28) we have

$$F(\theta)^T p \le c|h(\theta)| \left(-\lambda_1 + \lambda_2 \left(-\alpha + \frac{c|\rho|}{L^2 \tilde{f}} \right) \right) - \lambda_2 \|\nabla J(\theta)\|^2 \left(1 - f^2(\theta)\right) \quad (31)$$

for any θ such that $\|\nabla J(\theta)\| \le |\rho|c/L\tilde{f}$. Therefore choosing

$$\alpha > \max \left\{ \sup_{\theta \in B_{r^*}(\theta_c^*)} \frac{\|\nabla J(\theta)\|}{L}, \frac{c|\rho|}{L^2 \tilde{f}} \right\} \quad (32)$$

yields $F(\theta)^T p < 0$ for all $\theta \in \mathcal{C}_\rho \setminus \{\theta_c^*\}$.

So $F(\theta)^T p < 0$ for all $\theta \in \mathcal{C}_\rho \setminus \{\theta_c^*\}$, and $\mathcal{R}$ is forward invariant by Theorem 2. Note that α does not depend on $\bar{V}$, see (29) and (32), although α in general depends on ρ. This is very important, because otherwise the set $\mathcal{R}$ cannot be guaranteed to be made arbitrarily large (by means of choosing a large $\bar{V}$) in order to encompass any initial condition of (16).

For any $\Delta > 0$ and initial condition, $\theta(t_0) \in \bar{B}_\Delta(\theta_c^*)$, we are always able to find a $\mathcal{C}_\rho$ such that $\bar{B}_\Delta(\theta_c^*) \subseteq \mathcal{C}_\rho$. Then we choose $\alpha > 0$ as in the cases 1-3, as a function of ρ. Then we find a $\bar{V} > 0$ such that $\bar{B}_\Delta(\theta_c^*) \subseteq \mathcal{R}$, recalling that $\mathcal{R} = \{V_\alpha(\theta) \le \bar{V}\} \cap \mathcal{C}_\rho$. In the case 3b we use Assumption 5 to guarantee the existence of this $\bar{V}$. In Case 3a, any chosen set $\mathcal{C}_\rho$ is compact by assumption and $\bar{V}$ can be taken as the maximum of V_α on $\mathcal{C}_\rho$. So any trajectory of (16) is contained on $\mathcal{R}$. See Fig. 3 for the depiction of the sets.

Step 2: show $\max L_f V_\alpha(\theta) < 0$ for all $\theta \in \mathcal{R} \setminus \{\theta_c^*\}$. Now that we have demonstrated, using the generalized gradient of W, that $\mathcal{R}$ is a compact forward invariant set, we must show that $\max L_f V_\alpha(\theta) < 0$ for all $\theta \in \mathcal{R} \setminus \{\theta_c^*\}$ in order to claim asymptotic stability to θ_c^* using Theorem 1. Using Proposition 2, for any $p \in \partial V_\alpha(\theta)$, p takes the same form as that of $p \in \partial W(\theta)$ using the the three cases in Step 1. This is due to the similar form of V_α in (13) and W in (15) and the rule for computing the generalized gradients in Proposition 2. We showed in Step 1 that $F(\theta)^T p < 0$ in each case, therefore for the same α chosen in Step 1, $\max L_f V_\alpha(\theta) < 0$ for all $x \in \mathcal{R} \setminus \{\theta_c^*\}$. Therefore the dynamics (16) are asymptotically stable to the constrained minimizer θ_c^*. ■

The proof above relies on the invariance of $\mathcal{C}_\rho$ where we are guaranteed that for any unsafe initial condition $h(\theta(0)) = \rho$ we will never leave the set $\mathcal{C}_\rho$ and additionally, the boundaries of the sublevel set of Lyapunov function V_α. The choice of α is simple if sets $\mathcal{C}_\rho$ are compact as we know trajectories will never leave the bounded super-level sets of h, and the supremum in (29) exists. If $\mathcal{C}_\rho$ are not all compact, then we must have a radially unbounded V_α to argue that trajectories will never leave the bounded sublevel sets of V and some other set, $\mathcal{C}_\rho$.

The Lyapunov function parameter α depends on the size of the set of the initial conditions, because it is dependant on the invariant set $\mathcal{C}_\rho$. The parameter α used to prove stability of some initial condition $h(\theta(0)) = -1$ may be insufficient to prove the stability of the initial condition $h(\theta(0)) = -2$. But, a Lyapunov function with α chosen to prove the stability of the initial condition $h(\theta(0)) = -2$, will indeed be sufficient in showing the stability of $h(\theta(0)) = -1$.

This Lyapunov function has an interesting connection to recent literature: we note that a similar Lyapunov function for 'gradient flow' systems in a nonlinear programming setting was used to show local stability [4].

V. Safe Extremum Seeking

We introduce the Safe ES scheme for analytically unknown functions h and J in this section. We will then derive a 'reduced model' following the framework of [33]. The reduced model is essentially derived from (33)-(37), making the filtered estimate equations (34)-(37) algebraic under the appropriate time transformation and singular perturbation. The authors in [33] showed that the reduced model must be SPA stable in a for the algorithm itself in (33)-(37) to be SPA stable. We demonstrate that indeed SPA stability can be shown with the help of an invariant set from the boundaries of a sub level set of the Lyapunov function used in Lemma 3, $\{V_\alpha(\theta) \le \bar{V}\}$, and a sufficiently large set $\mathcal{C}_\rho$. Then we present our notion of practical safety and extend the results to dynamical systems.

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A. Algorithm Design

We introduce the algorithm:

$$\dot{\hat{\theta}} = k\omega_{\mathrm{f}}(-G_J + \min\{\|G_h\|^{-2}, M^+\} \max\{G_J^T G_h - c\eta_h, 0\} G_h) \tag{33}$$

$$\dot{G}_J = -\omega_{\mathrm{f}}(G_J - (J(\hat{\theta}(t) + S(t)) - \eta_J)M(t)) \tag{34}$$

$$\dot{\eta}_J = -\omega_{\mathrm{f}}(\eta_J - J(\hat{\theta}(t) + S(t))) \tag{35}$$

$$\dot{G}_h = -\omega_{\mathrm{f}}(G_h - (h(\hat{\theta}(t) + S(t)) - \eta_h)M(t)) \tag{36}$$

$$\dot{\eta}_h = -\omega_{\mathrm{f}}(\eta_h - h(\hat{\theta}(t) + S(t))) \tag{37}$$

where the state variables $\hat{\theta}, G_J, G_h \in \mathbb{R}^n$, $\eta_J, \eta_h \in \mathbb{R}$. The overall dimension of the system is $3n+2$. The map is evaluated at θ, defined by

$$\theta(t) \coloneqq \hat{\theta}(t) + S(t) \, . \tag{38}$$

The integer n denotes the number of parameters one wishes to optimize over. The design coefficients are $k, c, \delta, \omega_{\mathrm{f}}, M^+ \in \mathbb{R}_{>0}$. The perturbation signal S and demodulation signal M are given by

$$S(t) = a \left[\sin(\omega_1 t), \ \ldots \ , \sin(\omega_n t)\right]^T , \tag{39}$$

$$M(t) = \frac{2}{a} \left[\sin(\omega_1 t), \ \ldots \ , \sin(\omega_n t)\right]^T , \tag{40}$$

and contain additional design parameters $\omega_i, a \in \mathbb{R}_{>0}$.

B. Deriving the Reduced Model

First, we derive the reduce model. Defining

$$F_0(\xi) \coloneqq -\xi_1 + \min\{\|\xi_2\|^{-2}, M^+\} \max\{\xi_1^T \xi_2 - c\xi_4, 0\}\xi_3, \tag{41}$$

with

$$\xi^T \coloneqq [G_J^T, \eta_J, G_h^T, \eta_h], \tag{42}$$

$$\zeta^T(t, \theta, \xi, a) \coloneqq [(J(\theta) - \xi_2)M(t)^T, J(\theta), (h(\theta) - \xi_4)M(t)^T, h(\theta)], \tag{43}$$

we can rewrite (33) - (37) as

$$\dot{\hat{\theta}} = k\omega_{\mathrm{f}} F_0(\xi), \tag{44}$$

$$\dot{\xi} = -\omega_{\mathrm{f}}(\xi - \zeta(t, \theta, \xi, a)), \tag{45}$$

recalling $\theta = \hat{\theta} + S(t)$. Letting $\tau = \omega_{\mathrm{f}} t$, the system in the new time scale is

$$\frac{d\hat{\theta}}{d\tau} = kF_0(\xi), \tag{46}$$

$$\frac{d\xi}{d\tau} = -\left(\xi - \zeta\left(\frac{\tau}{\omega_{\mathrm{f}}}, \hat{\theta} + S\left(\frac{\tau}{\omega_{\mathrm{f}}}\right), \xi, a\right)\right). \tag{47}$$

We can take the average of the system (see Proposition 4 in Appendix) to compute

$$\frac{d\tilde{\theta}_{av}}{d\tau} = kF_0(\xi_{av}), \tag{48}$$

$$\frac{d\xi_{av}}{d\tau} = -\left(\xi_{av} - \mu(\hat{\theta}_{av}, a)\right), \tag{49}$$

$$D(\hat{\theta}_{av}) \coloneqq [\nabla J(\hat{\theta}_{av})^T, J(\hat{\theta}_{av}), \nabla h(\hat{\theta}_{av})^T, h(\hat{\theta}_{av})]^T, \tag{50}$$

$$\mu(\hat{\theta}_{av}, a) \coloneqq D(\hat{\theta}_{av}) + O(a). \tag{51}$$

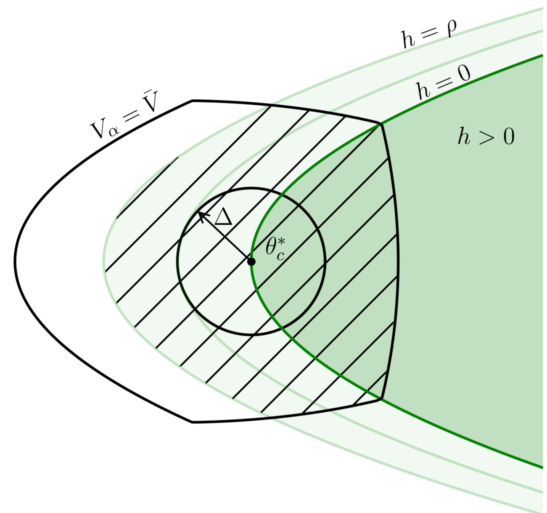

Fig. 3. Depiction of $\boldsymbol{V_\alpha = \bar{V}}$ subsuming the set of initial conditions in $\|\boldsymbol{\theta - \theta_c^*}\| \leq \boldsymbol{\Delta}$, where the shaded region marks $\boldsymbol{\mathcal{R}}$.

Making another time transformation $s = k\tau$ we have

$$\frac{d\hat{\theta}_{av}}{ds} = F_0(\xi_{av}), \tag{52}$$

$$k\frac{d\xi_{av}}{ds} = -\left(\xi_{av} - \mu(\hat{\theta}_{av}, a)\right). \tag{53}$$

Taking $k = 0$, we now derive the singularly perturbed (or "reduced") system with a quasi steady state

$$z_s \coloneqq \xi_{av} = \mu(\hat{\theta}_{av}, a). \tag{54}$$

Defining

$$y = \xi_{av} - \mu(\hat{\theta}_{av}, a), \tag{55}$$

the boundary layer system (with $\tau = s/k$) is

$$\frac{dy}{d\tau} = -\left(\xi_{av} - \mu(\hat{\theta}_{av}, a)\right) = -y. \tag{56}$$

The boundary layer system is UGAS uniformly in ξ_{av} and t_0. The reduced system is

$$\frac{d\theta_r}{ds} = F_0(\mu(\theta_r, a)) = F_0\left(D(\theta_r) + O(a)\right). \tag{57}$$

The reduced order model is an estimate of the original system (33)-(37) (in a different time scale) because we will later choose k to be small. A small k implies the slowness of the dynamics of (33) relative to (34)-(37). Therefore, the singular perturbation makes the filters in (34)-(37) fast, capturing the effect of a small k. Note that the singular perturbation is performed on an averaged version of the original system.

C. SPA Stability of the Reduced Model

The reduced system in (57) and (16) are similar with the distinct difference that the gradients and measurements of J and h are perturbed by a small $O(a)$ disturbance. We make the claim below that under a sufficiently small a, the reduced model is SPA stable. Using results from [33], this demonstrates that the algorithm itself in (33)-(37) to be SPA stable.

Lemma 4: Let Assumptions 1-3, 7 hold. Also, let either 4 and 5 hold or 6 hold. Then there exists a $\beta_\theta \in \mathcal{KL}$ such that: for any positive pair Δ, ν, there exists a $a^* > 0$ such that for all $a \in (0, a^*)$ the reduced system (57) satisfies

$$\|\theta(t) - \theta_c^*\| \leq \beta_\theta(\|\theta(0) - \theta_c^*\|, s) + \nu \tag{58}$$

for all $\|\theta(0) - \theta_c^*\| \leq \Delta$.

Proof: To show the SPA stability of (57) we make use of analogous arguments used in the proof of Lemma 3, assuming either Assumptions 4-5 or Assumption 6. Consider a set of initial conditions around the constrained minimizer $\theta_r(t_0) \in \bar{B}_\Delta(\theta_c^*)$, then there exist a larger ball $\bar{B}_\Delta(\theta_c^*) \subset \bar{B}_{\Delta^*}(\theta_c^*)$ with $0 < \Delta < \Delta^*$. There must exist a $\mathcal{C}_\rho$ such that $\bar{B}_\Delta(\theta_c^*) \subset \bar{B}_{\Delta^*}(\theta_c^*) \subset \mathcal{C}_\rho$. Then, there exists an $\alpha > 0$, corresponding to $V_\alpha(\theta)$, and a $\bar{V}$ such that such that $\mathcal{R} = \{V_\alpha(\theta) \leq \bar{V}\} \cap \mathcal{C}_\rho$ is compact and invariant for (16) any $0 < \bar{V} < \infty$ by Lemma 3. Therefore we choose a $\bar{V}$ such that $\bar{B}_\Delta(\theta_c^*) \subset \bar{B}_{\Delta^*}(\theta_c^*) \subset \mathcal{R}$. See the depiction in Fig. 3 - the set $\mathcal{R}$ subsumes the ball of initial conditions along with a slightly larger ball around the Δ ball.

Under Assumptions 1 - 2, there exists a $M^+, a_{max} > 0$ such that for all $a \in (0, a_{max})$ the disturbed dynamics in (57) can be written as

$$\frac{d\theta_r}{ds} = F_0(D(\theta_r) + O(a)) = F(\theta_r) + O(a) \tag{59}$$

for all $\theta_r \in \mathcal{R}$, Lemma 5 in the appendix.

In other words, if one chooses a small enough and M^+ large enough, then the reduced dynamics (57) are simply the exact dynamics in (16) with an additive $O(a)$ disturbance. Furthermore, one can make the set $\mathcal{R}$ forward invariant with sufficiently small a under the same arguments used in Step 1 in the proof of Lemma 3. The vector $F(\theta_r) + O(a)$ lies in $\mathcal{T}_\mathcal{R}(\theta_r)$ if $(F(\theta_r) + O(a))^T p < 0$ for each θ_r on $\mathrm{bd}(\mathcal{R})$. Since $F(\theta_r)^T p < 0$ on this compact region (Lemma 3) there exists a sufficiently small a that that $(F(\theta_r) + O(a))^T p < 0$ for $\theta \in \mathrm{bd}(\mathcal{R})$. This proves semiglobal stability of (57). To achieve the 'practical' part of the SPA stability of (57), a further reduction of a is needed and an analogous argument to that used in Step 2 in the proof of Lemma 3. Recall for $p \in \partial V_\alpha(\theta_r)$ we have that $F(\theta_r)^T p < 0$ and we are interested in the negativity of $F(\theta_r)^T p + O(a)^T p < 0$. For any $\delta > 0$ ball, $\bar{B}_\delta(\theta_c^*)$, there exists a $\bar{V}_\delta > 0$ such that $\{V_\alpha(\theta_r) \leq \bar{V}_\delta\} \subseteq \bar{B}_\delta(\theta_c^*)$ and a sufficiently small $a > 0$ such that trajectories of (57) converge to the set $\{V_\alpha(\theta_r) \leq \bar{V}_\delta\}$ by Theorem 2. This can be accomplished, with $\mathcal{R}' = \mathcal{R} \backslash \{V_\alpha(\theta_r) \leq \bar{V}_\delta\}$ by choosing a such that $\|O(a)^T p\| < |C|$ for $\theta_r \in \mathrm{bd}(\mathcal{R})$ where C is the maximum of $F(\theta_r)^T p$ on the set $\mathcal{R}'$. Therefore (57) is SPA stable for small a. ■

D. Main Results in Static Plants

We make the conclusion below with $z = \xi - \mu(\tilde{\theta}, a)$ and $\tilde{\theta} = \hat{\theta} - \theta_c^*$, guaranteeing practical asymptotic stabilization—semiglobally.

Theorem 3 (Semiglobal Practical Asymptotic Stability): Let Assumptions 1-3, 7 hold. Also, let either 4 and 5 hold or 6 hold. Then there exists $\beta_\theta, \beta_\xi \in \mathcal{KL}$ such that: for any positive pair (Δ, ν) there exist $M^+, \omega_\mathrm{f}^*, a^* > 0$, such that for any $\omega_\mathrm{f} \in (0, \omega_\mathrm{f}^*)$, $a \in (0, a^*)$, there exists $k^*(a) > 0$ such that for any $k \in (0, k^*(a))$ the solutions to (44)-(45) satisfy

$$\|\tilde{\theta}(t)\| \leq \beta_\theta\left(\|\tilde{\theta}(t_0)\|, k \cdot \omega_\mathrm{f} \cdot (t - t_0)\right) + \nu, \tag{60}$$

$$\|z(t)\| \leq \beta_\xi\left(\|z(t_0)\|, \omega_\mathrm{f} \cdot (t - t_0)\right) + \nu, \tag{61}$$

for all $\|(\tilde{\theta}(t_0), z(t_0))\| \leq \Delta$, and all $t \geq t_0 \geq 0$.

Sketch of proof: the proof of Theorem 3 follows the proof of Theorem 1 in [33] - we will summarize it this paragraph using terminology from [47]. The reduced system (57) is SPA stable in a uniformly in small a, demonstrated in Lemma 4. Using Lemma 1 in [46], this implies the average system (48) - (49) is SPA stable in k, a. This further implies the original system (46) - (47) is SPA stable in k, a, ω_f uniformly in k, a, ω_f in time scale s. Performing the appropriate transformations from s to t we achieve the result of Theorem 3.

Clarifying some key points: this argument uses the two results, Lemma 2 in [47] and Lemma 1 in [46], which are fundamentally based on Theorem 1 in [49]. Theorem 1 in [49] states several important key facts: 1) There is a time scale separation (by a factor k) between the convergence of parameter $\tilde{\theta}$ and the convergence of the boundary layer variable z. 2) The class $\mathcal{KL}$ functions governing the convergence of the reduced model and boundary layer model are the same functions which govern the slow and fast states of the actual system (see Remarks 8 and 15 in [49] for the choice of measuring functions). 3) We use the same $\mathcal{KL}$ function to describe the convergence of an average system as that of the actual system, see [49, Section V, "Classical Averaging"].

Equation (61) tell us about convergence of estimated quantities $\xi(t)$ to their exact values $D(\tilde{\theta}(t) + \theta_c^*)$. Using (51) we can write

$$e := \xi - D(\tilde{\theta} + \theta_c^*), \tag{62}$$

$$\|e(t)\| \leq \beta_\xi(\|z(t_0)\|, \omega_\mathrm{f}(t - t_0)) + v + |O(a)|. \tag{63}$$

The variable e can be thought of as the error of the estimated quantities - various measurements and gradients of the static maps. Note, the functions β_θ, β_ξ are independent of a, k, ω_f [47]. Because $\dot{\tilde{\theta}}$ is proportional to k (44), we can choose a k such that the change in $\tilde{\theta}(t)$ over some time interval can be small relative to the change in $e(t)$ over the same interval. The steady state offset in the error $e(t)$ can also be made small after some amount of time, with a small a. From Lemma 5 we know that there must be a small enough disturbance in the estimated quantities such that the dynamics can be written as the sum of the exact dynamics and a disturbance, but only after the transient of β_ξ has been sufficiently diminished.

Therefore we make the following claim encompassing 1) practical convergence to the safe set—semiglobally

2) practical safety for all time—semiglobally—if starting in the safe set. These two claims, implied by (64), demonstrate that an exponential bound on h in time is achieved for any sign of $h(\theta(t_0))$, which cannot be merely inferred directly from the design coefficient bounds given in Theorem 3.

Theorem 4: (Semiglobal Practical Convergence to Safe Set and Semiglobal Practical Safety) Let Assumptions 1–3, 7 hold. Also, let either 4 and 5 hold or 6 hold. For any $\Delta > 0$ and any $\delta > 0$: there exists $M^+, a^{**}, \omega_{\mathrm{f}}^{**} > 0$ such that for any $a \in (0, a^{**})$, $\omega_{\mathrm{f}} \in (0, \omega_{\mathrm{f}}^{**})$, there exists $k^{**}(a) > 0$ such that for any $k \in (0, k^{**}(a))$,

$$h(\theta(t)) \geq h(\theta(t_0)) e^{-ck\omega_{\mathrm{f}}(t-t_0)} + O(\delta) \tag{64}$$

for all $t \in [t_0, \infty]$ and there exists a $T \in [t_0, \infty]$ such that

$$h(\hat{\theta}(t)) \geq h(\theta(t_0)) - \delta \qquad \text{for } t_0 \leq t \leq T, \tag{65}$$

$$h(\hat{\theta}(t)) \geq h(\theta(t_0)) e^{-k\omega_{\mathrm{f}} c(t-T)} - \delta \qquad \text{for } T \leq t < \infty \tag{66}$$

for all $\|[\tilde{\theta}(t_0)^T, z(t_0)^T]\| \leq \Delta$.

Proof: Step 1: show for appropriate ranges of design coefficients, there exists a $T > 0$ such that for $t \geq T$, the dynamics of $\hat{\theta}$ simplify.

Consider (44)-(45) written as,

$$\dot{\hat{\theta}} = k\omega_{\mathrm{f}} F_0 \left(D(\hat{\theta}) + e(t) \right), \tag{67}$$

$$\|e(t)\| \leq \beta_\xi(\|z(t_0)\|, \omega_{\mathrm{f}}(t - t_0)) + v + |O(a)|. \tag{68}$$

with the bounds (63) restated, for some $\Delta, \nu > 0$ using Theorem 3, which follows from the assumptions. Theorem 3 states that there exist ranges of coefficients of $M^+, a, \omega_{\mathrm{f}}, k$ achieving the bound above, as well as the existence of an invariant set $\mathcal{R}$ which contains the set of initial conditions. Furthermore, there exists an ϵ^* such that for all ϵ, we can choose a $\nu > 0$ and restrict a to a smaller range such that $\nu + |O(a)| < \epsilon$ in (68), for $0 < \epsilon < \epsilon^*$ by Lemma 5.

In summary, for an appropriate ν and a (and appropriate ranges for the other design coefficients), there exist a time $T > 0$, after which (67) can be written as

$$\dot{\hat{\theta}} = k\omega_{\mathrm{f}} \left(F(\hat{\theta}) + r(t) \right) \tag{69}$$

where $r(t) = O(\nu + a) = O(\epsilon)$.

We will summarize the contents of Step 1 in this paragraph. After a time $T > 0$, the dynamics of the parameter can be written as the sum of the 'exact' gradient based scheme and a time dependent disturbance. The time $T > 0$ depends on the first choosing v (by Theorem 3, this choice defines the ranges of coefficients) and then restricting a further. The time $T > 0$ represents the time during which the estimated quantities' error below a threshold $\epsilon^* > 0$, which exists by Lemma 5.

Step 2: construct an approximate safe inequality for $T \leq t \leq \infty$.

For any $\delta > 0$, take ν and a, satisfying the condition $\nu + |O(a)| < \epsilon$ in Step 1, and sufficiently small to satisfy $\|r(t)\| < \delta c/b$ because $r(t) = O(\nu + a)$. The quantity $b > 0$ is the supremum of $\|\nabla h(\hat{\theta})\|$ on $\mathcal{R}$. From the result of Step 1, there exists a $T > 0$ such that (69) is true.

Next, compute $\dot{h} + ck\omega_{\mathrm{f}} h$:

$$\dot{h}(\hat{\theta}) + ck\omega_{\mathrm{f}} h(\hat{\theta}) = k\omega_{\mathrm{f}}(F(\hat{\theta})^T \nabla h(\hat{\theta}) + ch(\hat{\theta}) + r(t)^T \nabla h(\hat{\theta})), \tag{70}$$

$$\dot{h}(\hat{\theta}) + ck\omega_{\mathrm{f}} h(\hat{\theta}) \geq -\delta ck\omega_{\mathrm{f}}, \tag{71}$$

because $F(\hat{\theta})^T \nabla h(\hat{\theta}) + ch(\hat{\theta}) \geq 0$ by Proposition 3 and because $r(t)^T \nabla h(\hat{\theta}) \geq -\delta c$.

The approximate safety inequality $\dot{h} + ck\omega_{\mathrm{f}} h + \delta ck\omega_{\mathrm{f}} \geq 0$ implies, by the comparison principle, for $T \leq t < \infty$,

$$h(\hat{\theta}(t)) \geq (h(\hat{\theta}(T)) + \delta) e^{-k\omega_{\mathrm{f}} c(t-T)} - \delta. \tag{72}$$

Step 3: summarize and refine bounds.

For the period of time $t_0 \leq t \leq T$, one can find a range of k, sufficiently small, to achieve an arbitrarily small change the parameter (without affecting $T > 0$). Using the range of k given by Theorem 3, take k small such that

$$h(\hat{\theta}(t)) \geq h(\theta(t_0)) - \delta, \tag{73}$$

for $t_0 \leq t \leq T$. Using $h(\hat{\theta}(T)) + \delta \geq h(\theta(t_0))$ which follows from above, we summarize the bounds for all $t > 0$:

$$h(\hat{\theta}(t)) \geq h(\theta(t_0)) - \delta \qquad \text{for } t_0 \leq t \leq T, \tag{74}$$

$$h(\hat{\theta}(t)) \geq h(\theta(t_0)) e^{-k\omega_{\mathrm{f}} c(t-T)} - \delta \qquad \text{for } T \leq t < \infty \tag{75}$$

Already, one can simply stop here and present the result as is. In summary, for any $\delta > 0$, one achieves the bounds above by means of further restriction of the design coefficients that are given in Theorem 3.

Step 5: when $h(\theta(t_0)) \geq 0$, bound $h(\hat{\theta}(t))$ with a single inequality.

Consider first (74). Because $e^{-k\omega_{\mathrm{f}} c(t-t_0)} \leq 1$ on $t_0 \leq t \leq T$, we multiply the term $h(\theta(t_0))$ by $e^{-k\omega_{\mathrm{f}} c(t-t_0)}$ in (74) to yield

$$h(\hat{\theta}(t)) \geq h(\theta(t_0)) e^{-k\omega_{\mathrm{f}} c(t-t_0)} - \delta \text{ for } t_0 \leq t \leq T. \tag{76}$$

Next, consider (75), and using the fact that $e^{-k\omega_{\mathrm{f}} c(t_0-T)} \leq 1$, we have the following, similarly:

$$h(\hat{\theta}(t)) \geq h(\theta(t_0)) e^{-k\omega_{\mathrm{f}} c(t-t_0)} - \delta \text{ for } T \leq t < \infty. \tag{77}$$

Step 6: construct a single inequality which holds for $t_0 \leq t < \infty$ and all $h(\theta(t_0))$.

Consider the Taylor expansion of $e^{-k\omega_{\mathrm{f}} c(t-t_0)}$ about $k = 0$ for $t_0 \leq t \leq T$, which is $e^{-k\omega_{\mathrm{f}} c(t-t_0)} = 1 + O(k)$. Then, $h(\theta(t_0)) = h(\theta(t_0)) e^{-k\omega_{\mathrm{f}} c(t-t_0)} + O(k)$ and

$$h(\hat{\theta}(t)) \geq h(\theta(t_0)) e^{-k\omega_{\mathrm{f}} c(t-t_0)} - \delta + O(k) \text{ for } t_0 \leq t \leq T, \tag{78}$$

following from (76). Using a similar argument with $e^{-k\omega_{\mathrm{f}} c(t_0-T)} = 1 + O(k)$, we achieve

$$h(\hat{\theta}(t)) \geq h(\theta(t_0)) e^{-k\omega_{\mathrm{f}} c(t-t_0)} - \delta + O(k) \text{ for } T \leq t < \infty. \tag{79}$$

from (77). With an appropriately small k, we have

$$h(\hat{\theta}(t)) \geq h(\theta(t_0)) e^{-k\omega_{\mathrm{f}} c(t-t_0)} + O(\delta) \text{ for } t_0 \leq t < \infty. \tag{80}$$

■

Note that this result makes use of the fact that the time T is independent of k. Because the the estimator error convergence rate given by $\beta_\xi(\Delta, \omega_{\mathrm{f}}(t-t_0))$ is independent of k, there will be some finite time at which estimators are very close to their true values. During this finite time, we are able to shrink the change in h by restricting k. This result also relies on Theorem 3 and a specific choice of ν to select particular intervals on $a, \omega_{\mathrm{f}}, k$ which yield us the exponential safety result we seek. Therefore the intervals of $a, \omega_{\mathrm{f}}, k$ given in Theorem 4 are perhaps a more strict set of intervals than the ones given in Theorem 3 - if the user desires the type of safety given in (64).

Furthermore, this safety result is analogous to the statement on stability. The stability result says that for any set of initial conditions, one should be able to adjust $a, \omega_{\mathrm{f}}, k$ such that trajectories are ν-practically stable. The safety result says that for any set of initial conditions, one should be able to adjust $a, \omega_{\mathrm{f}}, k$ such that trajectories are δ-practically stable.

E. Main Results for Dynamical Systems

We follow a similar outline of the transformations and problem formulation given in [33]. Consider the dynamic system

$$\dot{x} = f(x, u), \qquad y_h = g_h(x), \qquad y_J = g_J(x), \tag{81}$$

where $f : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$ and $g_h, g_J : \mathbb{R}^n \to \mathbb{R}$. We will take y_J as our signal of performance and y_h as a measure of safety. Assume there exist the control law parameterized by θ as

$$u = \beta(x, \theta), \tag{82}$$

where $\theta \in \mathbb{R}^n$. The closed loop system is

$$\dot{x} = f(x, \beta(x, \theta)). \tag{83}$$

Assumption 8: There exists a function $l : \mathbb{R}^n \to \mathbb{R}^n$ such that $f(x, \beta(x, \theta)) = 0$ if and only if $x = l(\theta)$. And, for each $\theta \in \mathbb{R}^n$, the equilibrium $x = l(\theta)$ of (83) is globally asymptotically stable, uniformly in θ.

Under a singular perturbation and time scale transformation, the performance map will take the form $J(\cdot) = g_J(l(\cdot))$ and the barrier function $h(\cdot) = g_h(l(\cdot))$. We allow the perturbation frequencies ω_i to be scaled by a parameter ω_s. This is necessary to achieve results for dynamical systems, as the perturbation frequencies must be adjusted to be slower than the plant. Set the ith component of the perturbation and demodulation signals as $\omega_s t$ as $S_i(\omega_s t) = a\sin(\omega_i \omega_s t),\ M_i(\omega_s t) = (2/a)\sin(\omega_i\omega_s t)$.

We must redefine ζ using the measurements of safety and performance as y_h, y_J.

$$\begin{aligned}\zeta^T(\omega_s t, \theta, \xi, a) := [&(y_J - \xi_2)M(\omega_s t)^T, y_J,\\ &(y_h - \xi_4)M(\omega_s t)^T, y_h].\end{aligned} \tag{84}$$

The definition of ξ and F_0 are given by (42) and (41). Under the transformation $\theta = \hat{\theta} + S(\omega_s t)$ we have the proposed scheme for dynamical systems as

$$\begin{aligned}\dot{x} &= f(x, \beta(x, \hat{\theta} + S(\omega_s t))),\\ \dot{\hat{\theta}} &= k\omega_{\mathrm{f}}\omega_s F_0(\xi),\\ \dot{\xi} &= -\omega_{\mathrm{f}}\omega_s(\xi - \zeta(\omega_s t, \hat{\theta} + S(\omega_s t), \xi, a)).\end{aligned} \tag{85}$$

Letting $\sigma = \omega_s t$ in the new time scale we write the system as

$$\begin{aligned}\omega_s \frac{dx}{d\sigma} &= f(x, \beta(x, \hat{\theta} + S(\sigma))),\\ \frac{d\hat{\theta}}{d\sigma} &= k\omega_{\mathrm{f}} F_0(\xi),\\ \frac{d\xi}{d\sigma} &= -\omega_{\mathrm{f}}(\xi - \zeta(\sigma, \hat{\theta} + S(\sigma), \xi, a)).\end{aligned} \tag{86}$$

We construct a reduced system by letting $\omega_s = 0$ and taking $x = l(\hat{\theta} + S(\sigma))$. Using the time scale $\tau = \sigma\omega_{\mathrm{f}}$:

$$\begin{aligned}\frac{d\theta_s}{d\tau} &= kF_0(\xi_s),\\ \frac{d\xi_s}{d\tau} &= -\left(\xi_s - \zeta\left(\frac{\tau}{\omega_{\mathrm{f}}}, \theta_s + S\left(\frac{\tau}{\omega_{\mathrm{f}}}\right), \xi_s, a\right)\right).\end{aligned} \tag{87}$$

Also, we have that $y_h = g_h(l(\hat{\theta} + S(\tau/\omega_{\mathrm{f}})))$ and $y_J = g_J(l(\hat{\theta} + S(\tau/\omega_{\mathrm{f}})))$ appearing as terms in ζ.

With $J(\theta) := g_J(l(\theta))$ and $h(\theta) := g_h(l(\theta))$, the system (87) is the same as (46) - (47). The system was demonstrated to be SPA stable. Therefore we make the following conclusion based on similar arguments using averaging and singular perturbation results in Theorem 3, and the safety result in Theorem 4. Let $z_1(t) = \xi(t) - \mu(\hat{\theta}(t), a)$, $z_2(t) = x(t) - l(\hat{\theta}(t) + S(\omega_s t))$, $\tilde{\theta}(t) = \hat{\theta}(t) - \theta_c^*$.

Theorem 5 (Stability and Safety for Dynamical Systems): With $J(\theta) := g_J(l(\theta))$ and $h(\theta) := g_h(l(\theta))$ let Assumptions 1-3, 7-8 hold. Also, let either 4 and 5 hold or 6 hold. Then there exist $\beta_\theta, \beta_\xi, \beta_x \in \mathcal{KL}$ such that for any positive triple (Δ, ν, δ) there exists $a^* > 0$ and $\omega_{\mathrm{f}}^* > 0$ such that for any $a \in (0, a^*)$ and $\omega_{\mathrm{f}} \in (0, \omega_f^*)$ there exists $k^*(a) > 0$ such that for any $k \in (0, k^*(a))$, there exists $\omega_s^*(a, \omega_{\mathrm{f}}, k) > 0$ such that for any $\omega_s \in (0, \omega_s^*(a, \omega_{\mathrm{f}}, k))$ the solutions to (85) satisfy

$$\|\tilde{\theta}(t)\| \le \beta_\theta\left(\|\tilde{\theta}(t_0)\|, k\omega_{\mathrm{f}}\omega_s(t-t_0)\right) + \nu \tag{88}$$

$$\|z_1(t)\| \le \beta_\xi\left(\|z_1(t_0)\|, \omega_{\mathrm{f}}\omega_s(t-t_0)\right) + \nu \tag{89}$$

$$\|z_2(t)\| \le \beta_x\left(\|z_2(t_0)\|, (t-t_0)\right) + \nu \tag{90}$$

and

$$h(\theta(t)) \ge h(\theta(t_0))e^{-ck\omega_{\mathrm{f}}\omega_s(t-t_0)} + O(\delta) \tag{91}$$

for all $\|(\tilde{\theta}(t_0), z_1(t_0), z_2(t_0))\| \le \Delta$, and all $t \ge t_0 \ge 0$. The proof of (88) - (90) follows that of [33]. The proof of (91) follows naturally using the arguments in Theorem 4 (the argument is made in exactly the same manner when considering the bounds (88) - (89) in time scale σ). The only difference in the result compared to the static case follows from the fact that the parameter dynamics are scaled by the factor $k\omega_{\mathrm{f}}\omega_s$ resulting in the time constant in (91) as $ck\omega_{\mathrm{f}}\omega_s$ instead of $ck\omega_{\mathrm{f}}$ as in (64).

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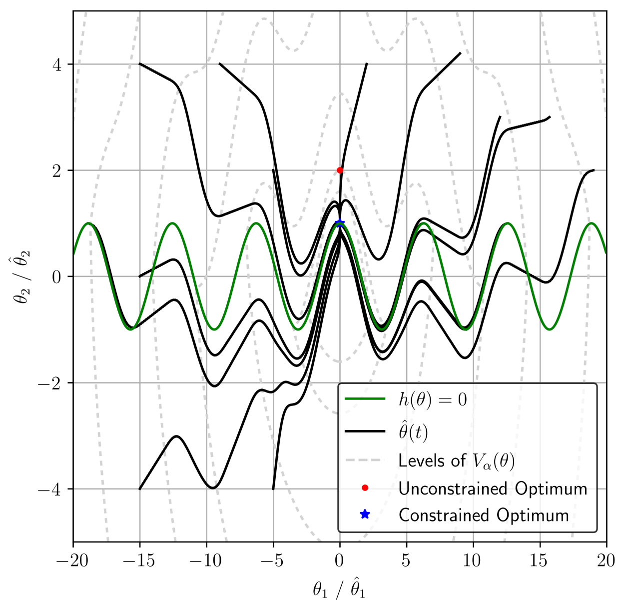

Fig. 4. Trajectories $\hat{\boldsymbol{\theta}}$ along with the safe boundary $\boldsymbol{h(\theta)=0}$ and level sets of $\boldsymbol{V_\alpha}$ with $\boldsymbol{\alpha=20}$. The safe set is below the curve $\boldsymbol{h(\theta)=0}$.

VI. Example

We present an example which demonstrates the capability of the Safe ES algorithm as well as the geometry of the Lyapunov function which proves the main stability result of this paper. The unknown CBF is $h(\theta)=\cos(\theta_1)-\theta_2$. This barrier function yields a nonconvex, semi-infinite safe set. The objective function is $J(\theta)=\theta_1^2+(\theta_2-2)^2$ with an unconstrained optimum at $\theta=(0,2)$.

One can check, for this problem, that our assumptions hold. First, we check Assumption 3 by computing the expression for $\nabla h(\theta)^T\nabla J(\theta)=\|\nabla h(\theta)\|\|\nabla J(\theta)\|$ for $h(\theta)=0$. After some algebra, we arrive at the following condition in θ_1:

$$q(\theta_1)=2\theta_1\sin(\theta_1)+2\cos(\theta_1)-4+ \\ 2\sqrt{\theta_1^2+(\cos(\theta_1)-2)^2}\sqrt{\sin(\theta_1)^2+1}=0, \quad (92)$$

which can be shown to have a unique solution at $\theta_1=0$ and a corresponding value $\theta_2=1$ which is the single constrained minimizer of J on $\mathcal{C}$.

Next, we check that Assumption 4 holds – perhaps one of the more unintuitive assumptions. After some algebra,

$$\frac{\nabla J(\theta)^T\nabla h(\theta)}{\|\nabla J(\theta)\|\|\nabla h(\theta)\|}=d_1(\theta_2)\frac{\theta_1+d_2(\theta_2)}{\sqrt{\theta_1^2+d_2^2(\theta_2)}}, \quad (93)$$

where $d_1(\theta_2)=-\sin(\theta_2)/\sqrt{\sin(\theta_2)^2+1}$ and $d_2(\theta_2)=\theta_2-2$. Recall we must check (93) is strictly less than 1 for $\rho\le h(\theta)\le 0$, for any $\rho<0$, far away from constrained minimizer of J on $\mathcal{C}$ - this implies θ_2 must be bounded, but θ_1 can be large. This means that $d_1(\theta_2),d_2(\theta_2)$ must be bounded with $d_2(\theta_2)$ having a more strict bound - approximately $\|d_2(\theta_2)\|\le 0.707$. As $\|\theta_1\|$ grows large it is clear than (93) tends towards 0.707, strictly less than 1. Other assumptions are straightforward to check.

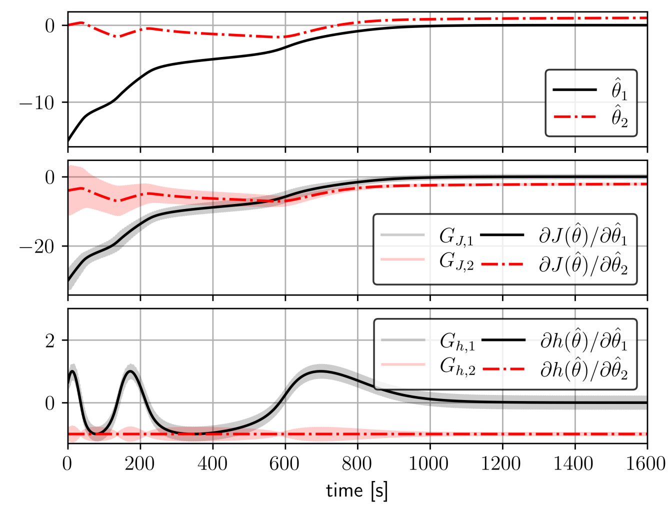

Fig. 5. Plot of $\hat{\boldsymbol{\theta}}(\boldsymbol{t})$ from $\hat{\boldsymbol{\theta}}(\boldsymbol{t_0})=\boldsymbol{(-15,0)}$ with estimates $\boldsymbol{G_J(t)}$ and $\boldsymbol{G_h(t)}$ compared with the true gradients $\boldsymbol{\nabla J(\hat{\theta})}$ and $\boldsymbol{\nabla h(\hat{\theta})}$.

In Fig. 4 we plot trajectories of the Safe ES algorithm, $\hat{\theta}(t)$, from various initial conditions. Design constants are chosen as $a=0.2$, $c=1$, $k=0.005$, $\omega_{\mathrm{f}}=0.5$, $M^+=10^4$, $\omega_1=10$, and $\omega_2=13$. We also show the shape of the level sets of a hypothesized Lyapunov function V_α for a value of $\alpha=20$. (Note: we prove that a sufficiently large α exists to demonstrate SPA stability, but it is complicated to check analytically if $\alpha=20$ is sufficient for a specific region of interest.) In the unsafe region, the contribution of h in V_α becomes noticeable in the wiggles of the level of V_α. This is because the practical safety result guarantees that trajectories “climb” the levels of h in order to escape to a safe region, and therefore h appears as a term in V_α in the unsafe region. In Fig. 5 we show that the gradient estimates G_J and G_h track their true values in time from the (unsafe) initial condition $\hat{\theta}(t_0)=(-15,0)$.

The trajectories shown are typical of safety filtered control, with $\hat{\theta}(t)$ slithering smoothly around unsafe regions. The geometric conditions on the static maps of the problem allow the constrained optimizer to be found, globally, with minor violations of safety along the way. Trajectories that start safe (or become safe) do not wander far out of the safe set for the rest of time, and all trajectories converge to a small region around the constrained optimum.

VII. Conclusion

This paper presents Safe ES, an ES controller designed to minimize an unknown objective function while ensuring the practical positivity of an unknown, yet measured CBF. In our approach, we add an approximation of a QP-based safety filter in the dynamics of the controller, advancing the current set of ES methodologies used to solve nonconvex constrained optimization problems. We use nonsmooth analysis tools and a key Lyapunov argument to help us achieve our main theoretical results. The notion of practical safety, akin to practical stability,

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uses known results which tell us that the estimates of the unknown gradients of the problem converge relatively fast (with respect to the parameter).

APPENDIX

A. Proof of Proposition 1

Proof: Let $J(\theta)=\frac{1}{2}\theta^T P\theta$ with $P\succ 0$ and $h(\theta)=h_1^T\theta+h_0$, with some constrained minimizer θ_c^*. We assume J has a global minimizer at 0 w.l.o.g. as we can always shift the coordinates to the form assumed.

Suppose Assumption 4 is not satisfied. Then $\exists\rho<0$ for all $r>0$ for all $f\in[0,1)$ such that

$$f<\hat{h}_1^T\frac{P\theta}{\|P\theta\|}\le 1, \tag{94}$$

for $\theta\in\{\rho\le h_1^T\theta+h_0\le 0\}\cap\{\|\theta-\theta_c^*\|\ge r\}$. Note $\hat{h}_1:=h_1/\|h_1\|$. Take $r=1$ and $f=1-\frac{1}{8}(\frac{\lambda_{\min}}{\lambda_{\max}})^2$ where $0<\lambda_{\min}\le\lambda_{\max}$ are the smallest and largest eigenvalues of P. For two unit vectors u,v we can write $\|u-v\|=\sqrt{2(1-u^Tv)}$. Therefore, with the inner product bound in (94) and the choice of f, we have

$$\left\|\hat{h}_1-\frac{P\theta}{\|P\theta\|}\right\|=\sqrt{2\left(1-\hat{h}_1^T\frac{P\theta}{\|P\theta\|}\right)}\in[0,\lambda_{\min}/2\lambda_{\max})\,. \tag{95}$$

The closeness relation between the two unit vectors $\hat{h}_1$ and $P\theta/\|P\theta\|$ implies that, for any point θ there is a unit vector $\hat{d}=\hat{d}(\theta)$ such that

$$\hat{h}_1=P\theta/\|P\theta\|+\epsilon\hat{d} \tag{96}$$

for $\epsilon\in[0,\lambda_{\min}/2\lambda_{\max})$.

Because $\theta\in\{\rho\le h_1^T\theta+h_0\le 0\}$ we have

$$\begin{aligned}\rho-h_0&\le\|h_1\|\|\theta\|\hat{\theta}^T\hat{h}_1\le -h_0,\\ \rho-h_0&\le\|h_1\|\|\theta\|\left(\frac{\hat{\theta}^TP\theta}{\|P\theta\|}+\epsilon\hat{\theta}^T\hat{d}\right)\le -h_0,\end{aligned} \tag{97}$$

substituting $\hat{h}_1$. The following quantity is strictly positive,

$$\frac{\hat{\theta}^TP\theta}{\|P\theta\|}+\epsilon\hat{\theta}^T\hat{d}>\frac{\lambda_{\min}}{\lambda_{\max}}-\frac{\lambda_{\min}}{2\lambda_{\max}}=\frac{\lambda_{\min}}{2\lambda_{\max}}>0. \tag{98}$$

Therefore (97) can not hold for sufficiently large $\|\theta\|$ because $\theta\in\{\|\theta-\theta_c^*\|\ge 1\}$. ∎

B. Proof of Lemma 1

Proof: Statement (1) follows from that fact that $\max\{a,0\}+\max\{b,0\}=\max\{a,b,a+b,0\}$ for any real a,b. In our case $a=-\alpha h(\theta)$ and $b=J(\theta)-J^*$. We also have that $\max\{-\alpha h(\theta),J(\theta)-J^*,-\alpha h(\theta)+J(\theta)-J^*,0\}=\max\{-\alpha h(\theta),J(\theta)-J^*,-\alpha h(\theta)+J(\theta)-J^*\}$ because $h(\theta)\ge 0$ implies $J(\theta)-J^*\ge 0$, by Assumption 1. By Proposition 2.3.12 in [12], V_α is regular because $h(\theta),J(\theta)$ each are regular because they are differentiable, and V_α can be expressed as a point-wise maximum of regular functions.

Statement (2a) follows directly from Proposition 2.1.2 and Theorem 2.4.7 in [12], if two conditions are met. The first condition is that the set $\mathcal{R}$ can be written as a sublevel set of a Lipschitz and regular function, as in $\mathcal{R}=\{W(\theta)\le 0\}$. The second condition is that for $W(\theta)=0$, $0\notin\partial W(\theta)$.

First, let $\tilde{J}(\theta)=J(\theta)-J^*$. We can express $\mathcal{R}$ as

$$\mathcal{R}=\{\theta:W(\theta)\le 0\}$$

where

$$W(\theta)=\max\{-h(\theta)+\rho^*,\tilde{J}(\theta)-\bar{V},\tilde{J}(\theta)-\bar{V}-\alpha h(\theta)\},$$

and $\rho^*=\max\{\rho,-\bar{V}/\alpha\}$. So, W is Lipschitz and regular by Proposition 2.3.12 [12].

Now we must show for $W(\theta)=0$, $0\notin\partial W(\theta)$. We consider θ lying in the three sets spanning $\mathbb{R}^n$. We use Proposition 2 to compute the generalized gradients of W.

Case 1) $h(\theta)>0$. Then $W(\theta)=\tilde{J}(\theta)-\bar{V}$ and $\partial W(\theta)=\{\nabla J(\theta)\}\ne\{0\}$ by Assumption 1.

Case 2) $h(\theta)=0$. Then $W(\theta)=\tilde{J}(\theta)-\bar{V}=\tilde{J}(\theta)-\bar{V}-\alpha h(\theta)$. So $\partial W(\theta)=\mathrm{co}\{\nabla J(\theta),\nabla J(\theta)-\alpha\nabla h(\theta)\}$. For some $p\in\partial W(\theta)$, $p=\nabla J(\theta)-\alpha\lambda\nabla h(\theta)$, for $0\le\lambda\le 1$. Because $\theta\ne\theta_c^*$, by Assumption 3 we have that $p\ne 0$.

Case 3) $h(\theta)<0$. So $\partial W(\theta)=\mathrm{co}\{-\nabla h(\theta),\nabla J(\theta)-\alpha\nabla h(\theta)\}$. For some $p\in\partial W(\theta)$, we have the expression $p=-\lambda_1\nabla h(\theta)+\lambda_2(-\alpha\nabla h(\theta)+\nabla J(\theta))$, with $0\le\lambda_1,\lambda_2\le 1$ and $\lambda_1+\lambda_2=1$.

If Assumption 6 holds, clearly $p\ne 0$ for sufficiently large α because $\|\nabla h(\theta)\|>L$ by Assumption 2. Choosing $\alpha>L^{-1}\max_{\theta\in\mathcal{C}_\rho}\|\nabla J(\theta)\|$, guarantees that $-\alpha\nabla h(\theta)+\nabla J(\theta)\ne 0$ and therefore $p\ne 0$.

If Assumptions 4 - 5 hold, then $\nabla h(\theta)\ne s\nabla J(\theta)$ for $s>0$ outside some ball around θ_c^*. We can choose $\alpha>L^{-1}\sup_{\theta\in B_{r^*}(\theta_c^*)}\|\nabla J(\theta)\|$. Then $p\ne 0$. This demonstrates Statement (2a) and Statement (2c).

Statement (2b): $\mathcal{R}$ is compact due to either Assumption 4- 5 holding, or 6 holding.

Now we prove $\{W(\theta)=0\}\equiv\mathrm{bd}(\mathcal{R})$. Step 1: show if $\theta\in\mathrm{bd}(\mathcal{R})$ then $\theta\in\{W(\theta)=0\}$. By contradiction, suppose $\theta\in\mathrm{bd}(\mathcal{R})$ and $\theta\in\{W(\theta)<0\}$. By continuity of W, there exists a $B_\epsilon(\theta)$ such that $W(y)<0$ for all $y\in B_\epsilon(\theta)$. Therefore $B_\epsilon(\theta)\subseteq\{W(\theta)<0\}$. But $B_\epsilon(\theta)\cap\mathcal{R}^c=B_\epsilon(\theta)\cap\{W(\theta)>0\}=\emptyset$, which contradicts $\theta\in\mathrm{bd}(\mathcal{R})$. Step 2: show if $\theta\in\{W(\theta)=0\}$ then $\theta\in\mathrm{bd}(\mathcal{R})$. By contradiction, suppose $\theta\in\{W(\theta)=0\}$ and $\theta\notin\mathrm{bd}(\mathcal{R})$. Then $\theta\notin\bar{\mathcal{R}}$ or $\theta\in\mathrm{Int}(\mathcal{R})$. If $\theta\notin\bar{\mathcal{R}}$ then $\theta\notin\mathcal{R}$ because $\mathcal{R}$ is closed, which implies $W(\theta)>0$ which is a contradiction. If $\theta\in\mathrm{Int}(\mathcal{R})$ then there exists a $B_\epsilon(\theta)$ such that $B_\epsilon(\theta)\subseteq\mathrm{Int}(\mathcal{R})\subseteq\mathcal{R}$. Since $\theta\in\{W(\theta)=0\}$ then $W(\theta)$ achieves a local maximum on $B_\epsilon(\theta)$ which implies $0\in\partial W(\theta)$ by Proposition 2.3.2 in [12], which is a contradiction as $0\notin\partial W(\theta)$ by Statement (2a). ∎

C. Lemma 5

We group the quantities as the function Q,

$$Q(\theta)^T:=[\nabla J(\theta)^T,\nabla h(\theta)^T,h(\theta)]. \tag{99}$$

representing (16) as a function of the estimated variables $F=F(Q(\theta))$. And with a small disturbance $w(t)^T=$

$[w_1(t)^T, w_2(t)^T, w_3(t)]$ we can consider a system, with disturbances to Q, written as

$$\dot{\theta} = F(Q(\theta) + w(t)). \tag{100}$$

Lemma 5 (Additive Disturbance): Under Assumptions 1 - 2, for any compact set Ω there exists a $M^+, \epsilon^* > 0$ such that for all $\epsilon \in (0, \epsilon^*)$ and $\|w\| < \epsilon$, the disturbed dynamics in (100) can be written as

$$F(Q(\theta) + w(t)) = -\nabla J(\theta) + \frac{\nabla h(\theta)}{\|\nabla h(\theta)\|^2} \max\{\nabla J(\theta)^T \nabla h(\theta) - ch(\theta), 0\} + O(\epsilon). \tag{101}$$

for all $\theta \in \Omega$.

Proof: Step 1 (Rewrite Dynamics, Grouping Terms): Note that terms $w(t), w_i(t) = O(\|w\|)$ for $i \in \{1, 2, 3\}$. We write the perturbed dynamics $F(Q(\theta) + w(t)) = F_w$ as

$$\begin{aligned} F_w =& -\nabla J(\theta) - w_1(t) + \min\{\|\nabla h(\theta) + w_2(t)\|^{-2}, M^+\}\cdot \\ &\max\{(\nabla J(\theta) + w_1(t))^T(\nabla h(\theta) + w_2(t)) - \\ &c(h(\theta) + w_3(t)), 0\}(\nabla h(\theta) + w_2(t)), \\ =& -\nabla J(\theta) + \min\{\|\nabla h(\theta) + w_2(t)\|^{-2}, M^+\}\cdot \\ &\max\{\nabla J(\theta)^T \nabla h(\theta) - ch(\theta), 0\}\nabla h(\theta) + O(\|w\|), \end{aligned} \tag{102}$$

where we group terms proportional to w_i as $O(\|w\|)$ and note that the max term is proportional (or zero) to its' argument.

Step 2 (Find Large Enough $\mathbf{M}^+$, Small Enough $\mathbf{w}$): Next, we find a suitably large M^+ such that the max term is not active when $\min\{\|\nabla h(\theta) + w_2(t)\|^{-2}, M^+\} = M^+$. First, Consider the infimum

$$G = \inf_{\theta \in \mathcal{S}} \|\nabla h(\theta)\| > 0 \tag{103}$$

$$\mathcal{S} = \Omega \cap \{\nabla J(\theta)^T \nabla h(\theta) - ch(\theta) \geq 0\}. \tag{104}$$

We have that $G > 0$ because for $\theta \in \{h(\theta) \leq 0\} \cap \mathcal{S}$, $\|\nabla h(\theta)\| > L > 0$ (Assumption 2) and for $\theta \in \mathcal{S} \setminus \{h(\theta) \leq 0\}$ we have that $\nabla J(\theta)^T \nabla h(\theta) > 0$ which implies $\|\nabla h(\theta)\| > 0$.

Now, we choose $\|w_2(t)\| < G/2$ and $M^+ > 4/G^2$. Therefore,

$$\frac{1}{\|\nabla h(\theta) + w_2(t)\|^2} \leq \frac{1}{(G/2)^2} = \frac{4}{G^2} < M^+ \tag{105}$$

for $\theta \in \mathcal{S}$. Taking $\epsilon^* = G/2$, we can express the dynamics without the min term as

$$F_w = -\nabla J(\theta) + \frac{\max\{\nabla J(\theta)^T \nabla h(\theta) - ch(\theta), 0\}}{\|\nabla h(\theta) + O(\epsilon)\|^2} \nabla h(\theta) + O(\epsilon), \tag{106}$$

for $0 < \epsilon < \epsilon^*$. Note also the bound for the term

$$\frac{1}{\|\nabla h(\theta) + O(\epsilon)\|^2} < M^+ \tag{107}$$

holding for $\theta \in \mathcal{S}$, for all $0 < \epsilon < \epsilon^*$. In (106), we have now also taken the whole vector $w(t)$ as ϵ-small for simplicity. Note we also could have chosen $\epsilon^* = \mu G$ (and the bound on $\|w_2(t)\|$) for any $\mu \in (0, 1)$. If taking μ to be larger than 1/2 (we took it as exactly 1/2), this would have forced a larger choice of M^+.

Step 3 (Find an $\mathbf{O}(\epsilon)$ Approximation of $\|\nabla \mathbf{h}(\theta) + \mathbf{O}(\epsilon)\|^{-\mathbf{2}}$): We now state two results. The first follows from the triangle inequality, and then squaring both sides of the resulting inequality:

$$\begin{aligned} &\|\nabla h(\theta) + O(\epsilon)\| \leq \|\nabla h(\theta)\| + \|O(\epsilon)\|, \\ \implies &\|\nabla h(\theta) + O(\epsilon)\|^2 \leq \|\nabla h(\theta)\|^2 + \|O(\epsilon)\|^2 + \\ &\qquad 2\|O(\epsilon)\|\|\nabla h(\theta)\|, \\ \implies &\|\nabla h(\theta) + O(\epsilon)\|^2 - \|\nabla h(\theta)\|^2 = O(\epsilon). \end{aligned} \tag{108}$$

The second next result uses the previous result, and determines a bound on $d(\theta)$, with

$$d(\theta) = \left\| \frac{1}{\|\nabla h(\theta) + O(\epsilon)\|^2} - \frac{1}{\|\nabla h(\theta)\|^2} \right\|. \tag{109}$$

We have,

$$d(\theta) = \left\| \frac{\|\nabla h(\theta)\|^2 - \|\nabla h(\theta) + O(\epsilon)\|^2}{\|\nabla h(\theta) + O(\epsilon)\|^2 \|\nabla h(\theta)\|^2} \right\|, \tag{110}$$

$$= \left\| \frac{\|\nabla h(\theta)\|^2 - \|\nabla h(\theta)\|^2 + O(\epsilon)}{\|\nabla h(\theta) + O(\epsilon)\|^2 \|\nabla h(\theta)\|^2} \right\|, \tag{111}$$

$$= \frac{\|O(\epsilon)\|}{\|\nabla h(\theta) + O(\epsilon)\|^2 \|\nabla h(\theta)\|^2},$$

$$\implies \frac{1}{\|\nabla h(\theta) + O(\epsilon)\|^2} - \frac{1}{\|\nabla h(\theta)\|^2} = O(\epsilon), \tag{112}$$

for $\theta \in \mathcal{S}$. We used the fact in the final line that $1/(\|\nabla h(\theta) + O(\epsilon)\|^2 \|\nabla h(\theta)\|^2) < M^+/G^2$ is bounded (see (107)) for $\theta \in \mathcal{S}$. We now use (112) to express $1/(\|\nabla h(\theta) + O(\epsilon)\|^2)$ in (106) as

$$\begin{aligned} F_w =& -\nabla J(\theta) + \max\{\nabla J(\theta)^T \nabla h(\theta) - ch(\theta), 0\}\cdot \\ &\left(\frac{1}{\|\nabla h(\theta)\|^2} + O(\epsilon)\right) \nabla h(\theta) + O(\epsilon), \end{aligned} \tag{113}$$

$$= -\nabla J(\theta) + \frac{\max\{\nabla J(\theta)^T \nabla h(\theta) - ch(\theta), 0\}}{\|\nabla h(\theta)\|^2} \nabla h(\theta) + O(\epsilon). \tag{114}$$

■

D. Averaging

Proposition 4 (Averaging in $\mathcal{C}^1$): Suppose $Q : \mathbb{R}^n \to \mathbb{R}$ is differentiable with locally Lipschitz Jacobian, then under Assumption 7 the following averages can be computed as

$$\frac{1}{\Pi} \int_0^\Pi Q(\theta + S(t)) M(t) dt = \nabla Q(\theta) + O(a), \tag{115}$$

$$\frac{1}{\Pi} \int_0^\Pi Q(\theta + S(t)) dt = Q(\theta) + O(a^2). \tag{116}$$

Proof: With $S_i(t) = a \sin(\omega_i' t)$ and $M_i(t) = \frac{2}{a} \sin(\omega_i' t)$, we take the period to be Π given as

$$\Pi = 2\pi \times \text{LCM}\left\{\frac{1}{\omega_i'}\right\}, \quad i \in \{1, 2, \ldots, n\}, \tag{117}$$

where LCM is the least common multiple. Consider

$$\frac{1}{\Pi}\int_0^{\Pi} S_i(t)M_j(t)dt = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \tag{118}$$

$$\frac{1}{\Pi}\int_0^{\Pi} S_i(t)S_j(t)dt = \begin{cases} \frac{a^2}{2} & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \tag{119}$$

and note that $Q(\theta + S(t))$ can be Taylor expanded [19, Theorem B.6] around θ as

$$Q(\theta + S(t)) = Q(\theta) + S(t)^T DQ(\theta) + O(a^2). \tag{120}$$

Now let us expand the LHS of (115) as

$$\frac{1}{\Pi}\int_0^{\Pi} Q(\theta + S(t))M(t)dt = \frac{1}{\Pi}\int_0^{\Pi} \Big(Q(\theta)M(t) + S(t)^T DQ(\theta)M(t) + O(a^2)M(t)\Big)dt \tag{121}$$

into three terms. The term $Q(\theta)M(t)$ averages to zero because M has zero average. The term $S(t)^T DQ(\theta)M(t)$ averages to $DQ(\theta)$ by (118). The last term can be written as $O(a)$ because $M \propto 1/a$. Therefore

$$\frac{1}{\Pi}\int_0^{\Pi} Q(\theta + S(t))M(t)dt = DQ(\theta) + O(a). \tag{122}$$

Because $S(t)$ averages to 0, expanding (116) we have

$$\frac{1}{\Pi}\int_0^{\Pi} Q(\theta + S(t))dt = \frac{1}{\Pi}\int_0^{\Pi} \Big(Q(\theta) + S(t)^T DQ(\theta) + O(a^2)\Big)dt \tag{123}$$

$$= Q(\theta) + O(a^2). \tag{124}$$

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